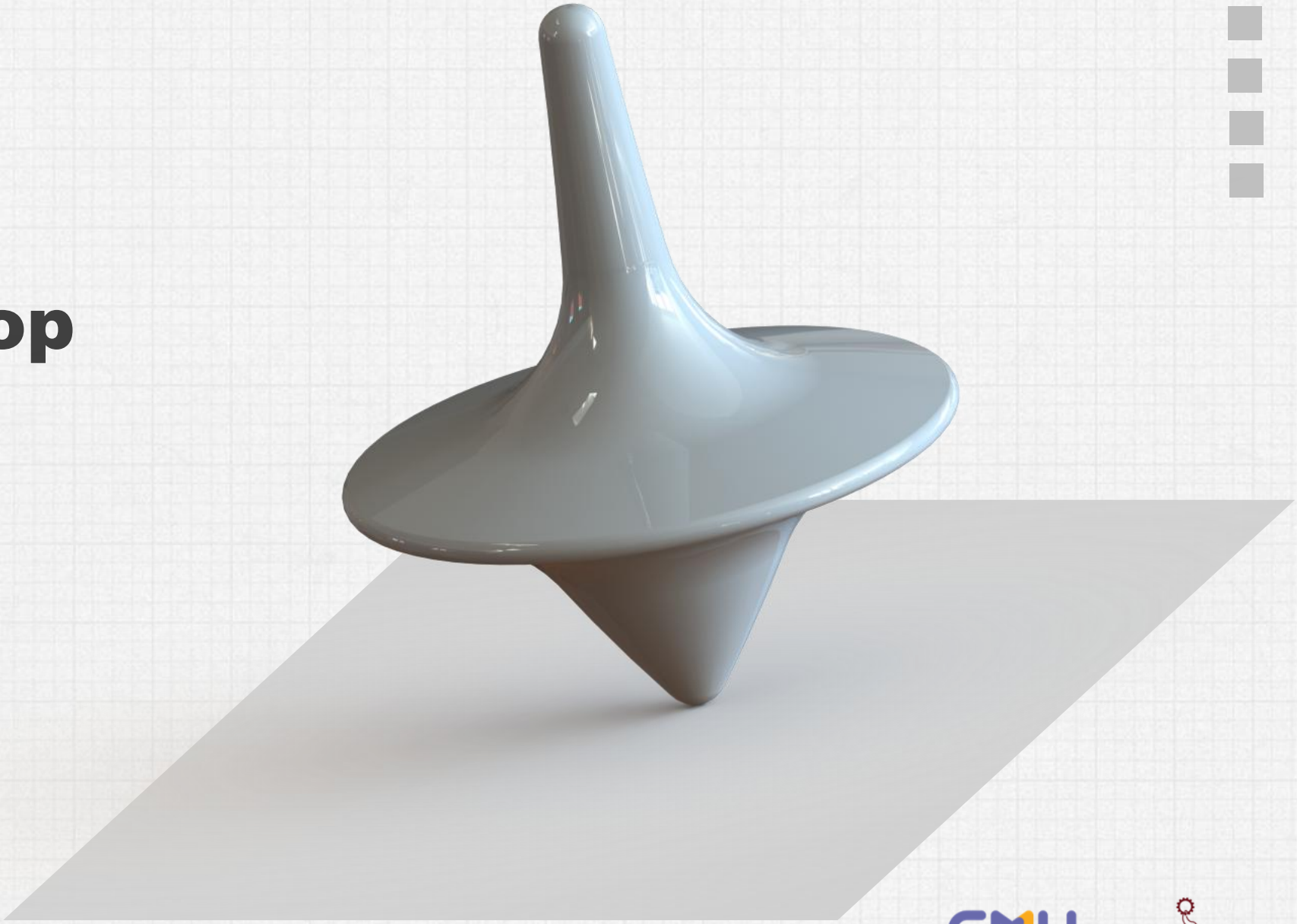
Analytical Dynamics

Final Project
Motion of a Top



Contents

- Description of Motion of a Top
- Derivation of Lagrange's EOM
- Numerical Simulation Setup
- Simulation Case and Result

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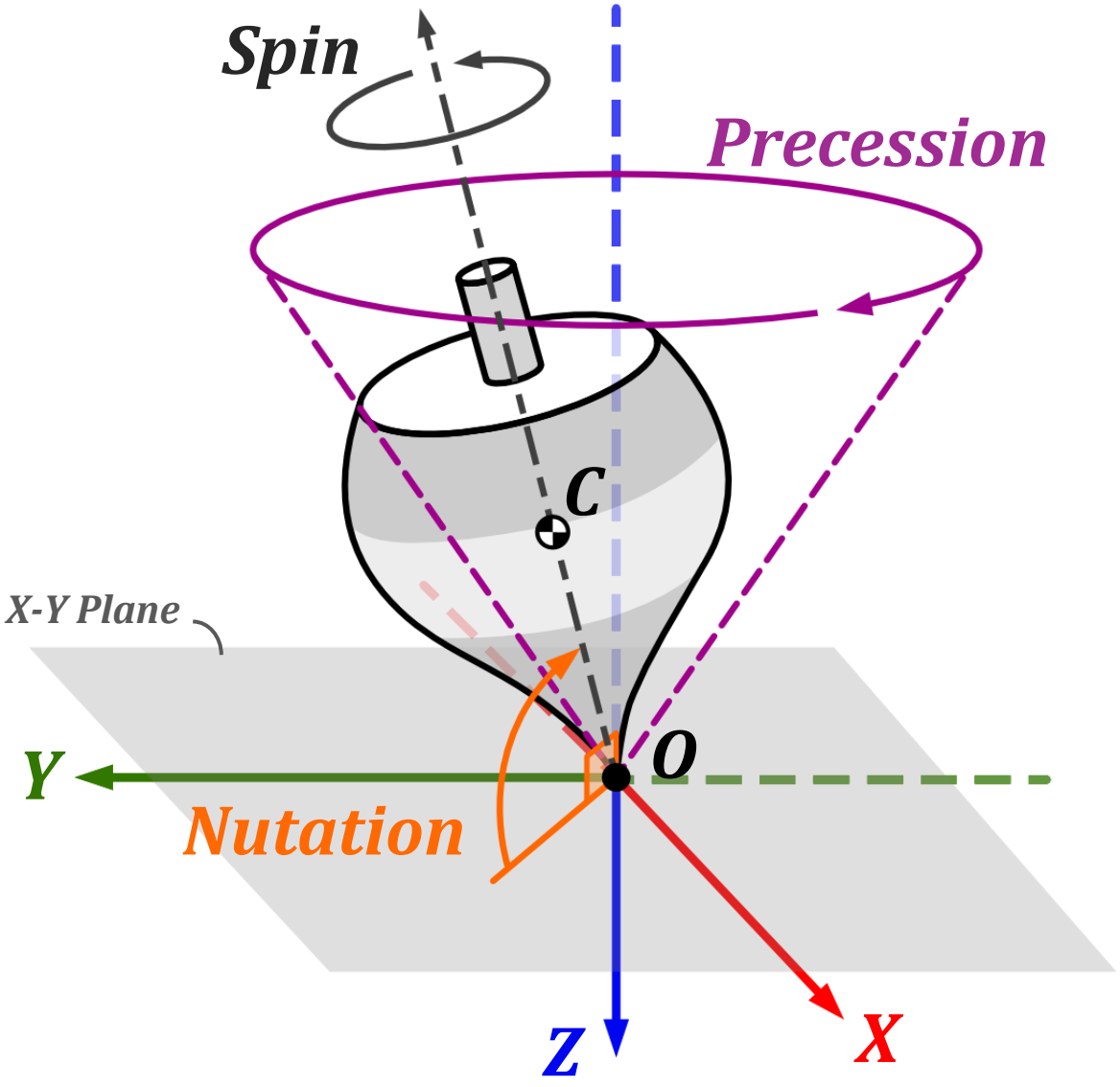
Description of Motion of a Top

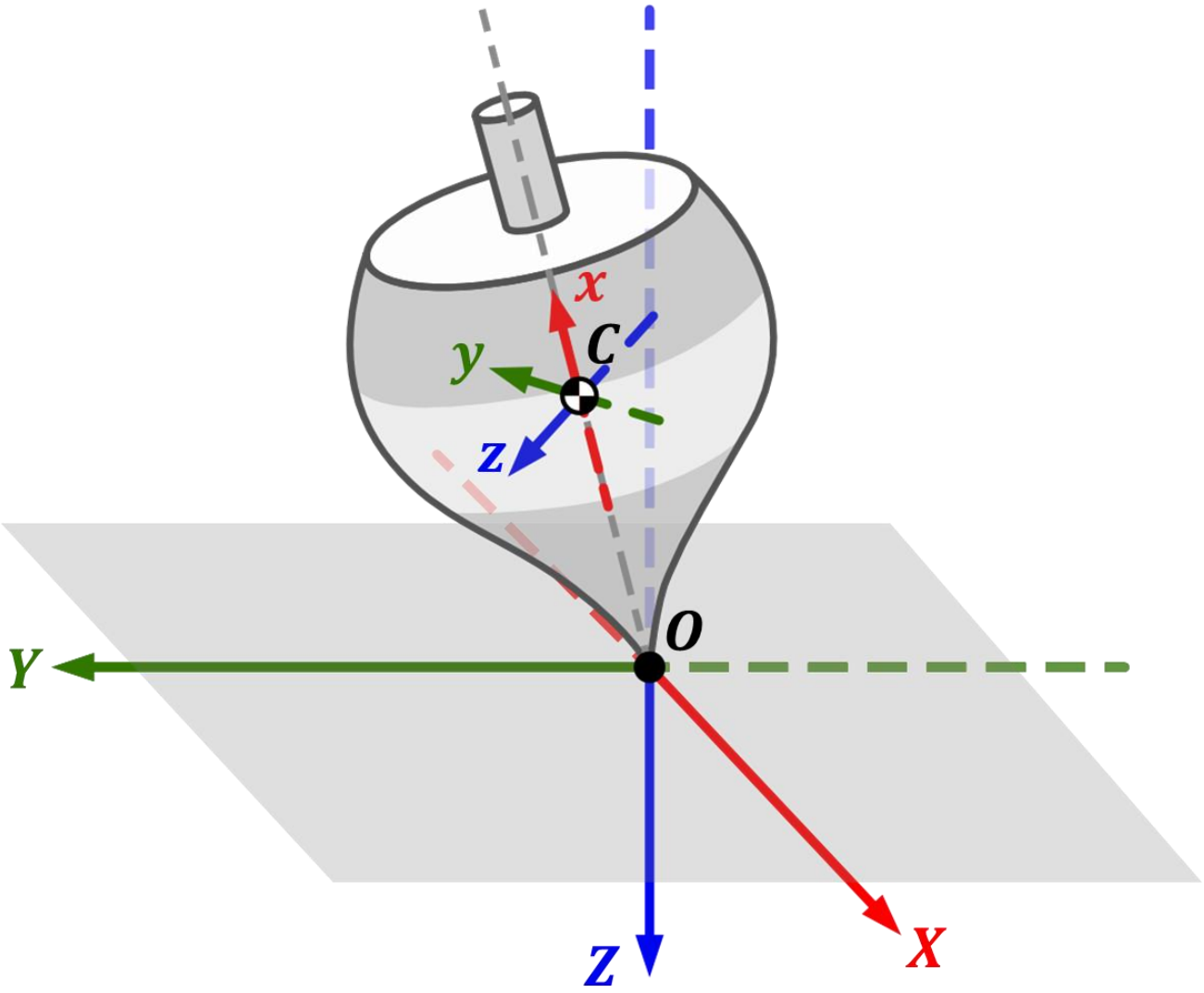
Precession
is the rotation of the top's axis around the fixed vertical axis

Nutation
is the tipping angle of the top's symmetry axis relative to the horizontal plane

Spin
is the top's rotation around its own axis of symmetry



DERIVATION OF LAGRANGE's EOM



Generalized Coordinate

3-DOF of motion

$$q_i \in \{\psi, \theta, \phi\}$$

Euler Angles

Reference Frames

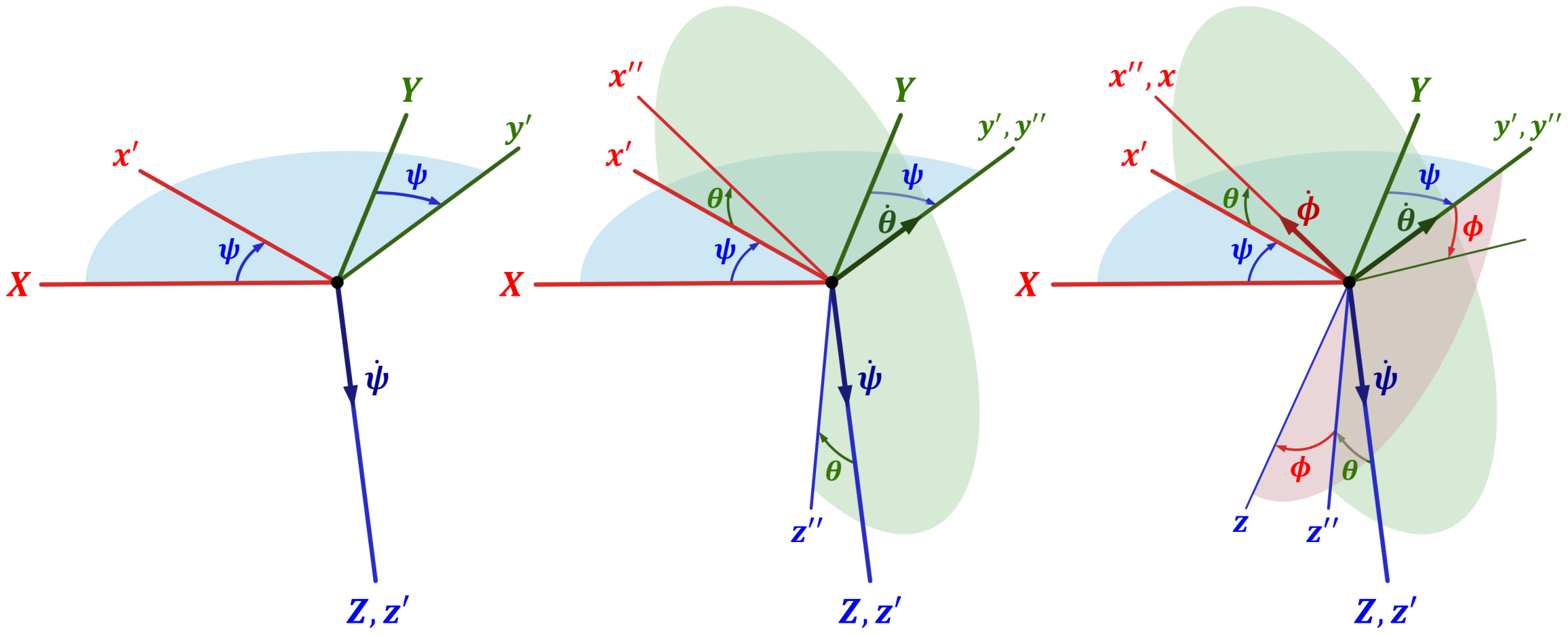
Body-Fixed Frame

(x, y, z)

Fixed Frame/Inertia Frame

(X, Y, Z)

Euler Angles Transformation



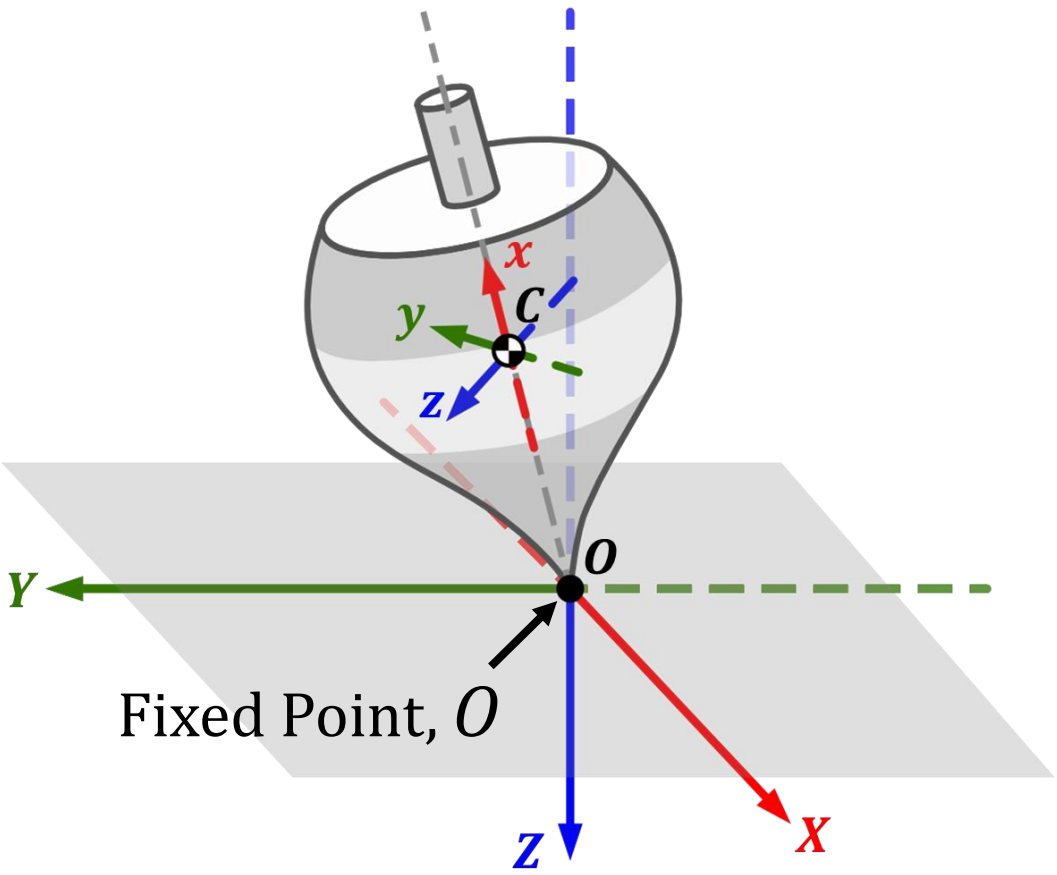
DERIVATION OF LAGRANGE's EOM

Kinetic Energy

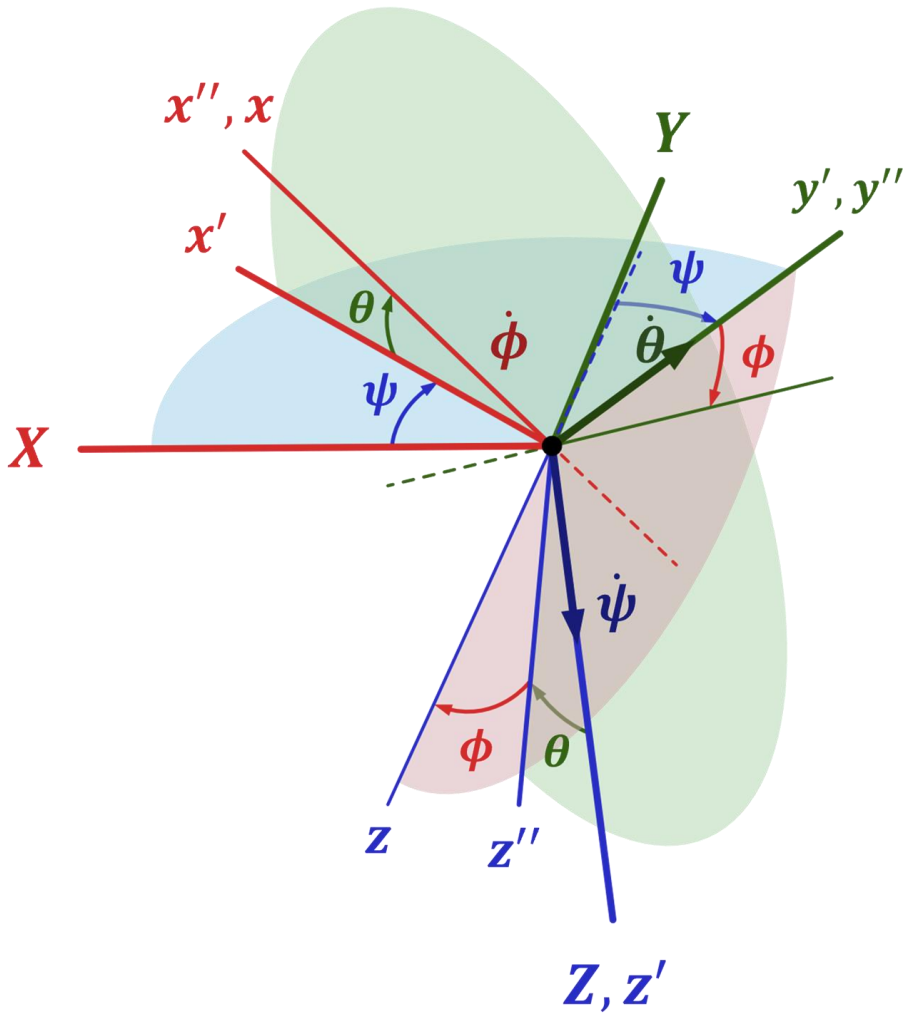
$$T_{total} = \cancelto{0}{T_{trans}} + T_{rot}$$

Translational Kinetic Energy

Rotational Kinetic Energy



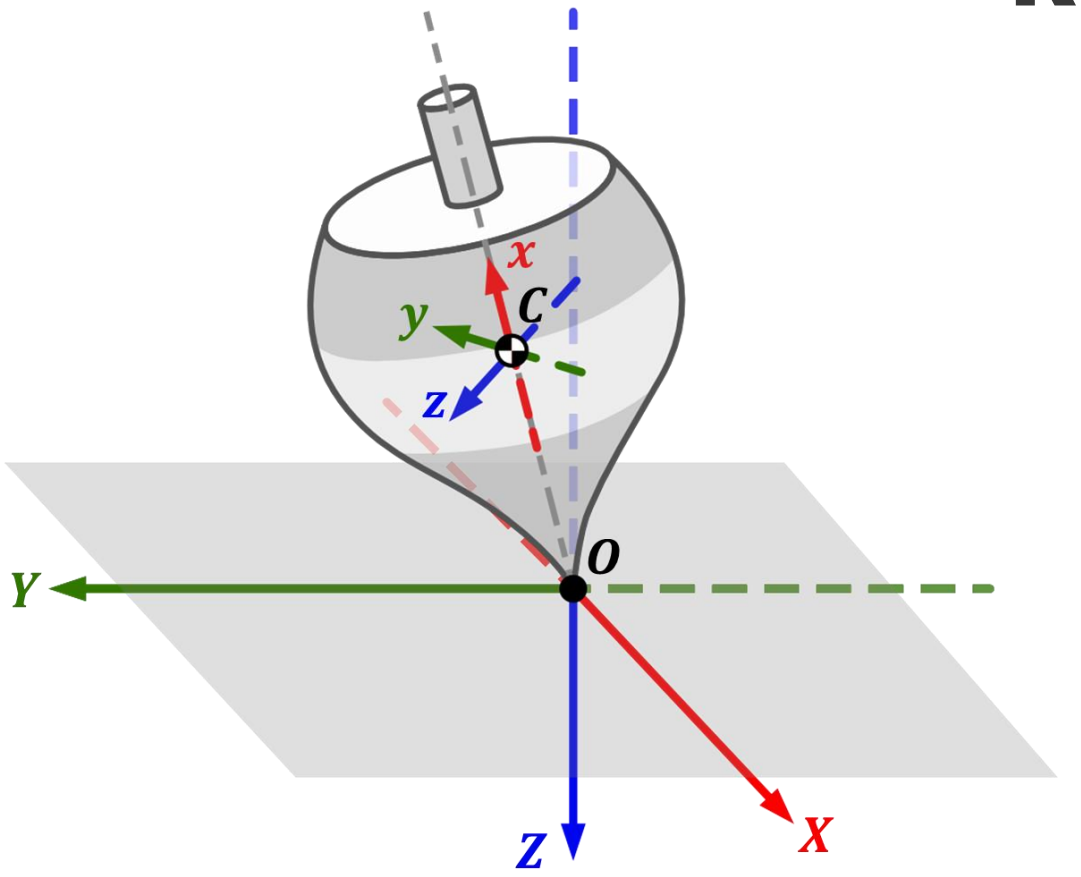
DERIVATION OF LAGRANGE's EOM



Rotational Kinetic Energy

$$T_{rot} = \frac{1}{2}\left(I_{xx}\omega_x^2 + I_{yy}\omega_y^2 + I_{zz}\omega_z^2\right)$$

DERIVATION OF LAGRANGE's EOM



Rotational Kinetic Energy

$$T_{rot} = \frac{1}{2}\left(I_{xx}\omega_x^2 + I_{yy}\omega_y^2 + I_{zz}\omega_z^2\right)$$

Assume the top is symmetry about the x-axis

Axial Moment of Inertia

$$I_{xx} = I_a$$

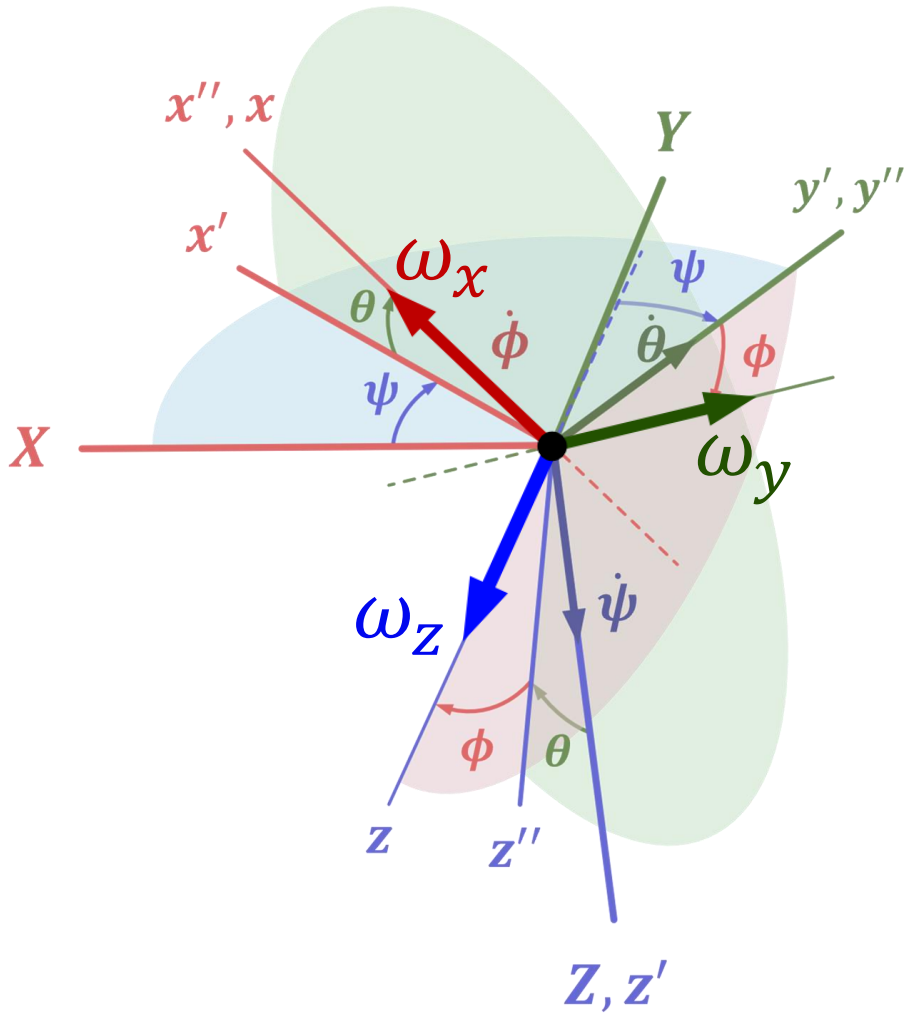
Transverse Moment of Inertia

$$I_{yy} = I_{zz} = I_t$$

$$\downarrow$$

$$T_{rot} = \frac{1}{2}\left(I_a\omega_x^2 + I_t\left(\omega_y^2 + \omega_z^2\right)\right)$$

DERIVATION OF LAGRANGE's EOM



Rotational Kinetic Energy

$$T_{rot} = \frac{1}{2}\left(I_a \omega_x^2 + I_t\left(\omega_y^2 + \omega_z^2\right)\right)$$

$$\boxed{\begin{aligned} \omega_x &= \dot{\phi} - \dot{\psi}\sin\theta \\ \omega_y &= \dot{\theta}\cos\phi + \dot{\psi}\cos\theta\sin\phi \\ \omega_z &= \dot{\psi}\cos\theta\cos\phi - \dot{\theta}\sin\phi \end{aligned}}$$

Total Spin Rate Ω

$$\omega_y^2 + \omega_z^2 = \left(\dot{\theta}\cos\phi + \dot{\psi}\cos\theta\sin\phi\right)^2 + \left(\dot{\psi}\cos\theta\cos\phi - \dot{\theta}\sin\phi\right)^2$$

$$= \dot{\theta}^2\cos^2\phi + 2\dot{\theta}\dot{\psi}\cos\phi\cos\theta\sin\phi + \dot{\psi}^2\cos^2\theta\sin^2\phi + \dot{\psi}^2\cos^2\theta\cos^2\phi - 2\dot{\theta}\dot{\psi}\cos\phi\cos\theta\sin\phi + \dot{\theta}^2\sin^2\phi$$

$$= \dot{\theta}^2(\cos^2\phi + \sin^2\phi) + \dot{\psi}^2\cos^2\theta\,(\sin^2\phi + \cos^2\phi)$$

Note: $\sin^2\phi + \cos^2\phi = 1$

$$\omega_y^2 + \omega_z^2 = \dot{\theta}^2 + \dot{\psi}^2\cos^2\theta$$

DERIVATION OF LAGRANGE's EOM

Total Kinetic Energy

$$T = \frac{1}{2}\left(I_a\left(\dot{\phi} - \dot{\psi}\sin\theta\right)^2 + I_t\left(\dot{\theta}^2 + \dot{\psi}^2\cos^2\theta\right)\right)$$

DERIVATION OF LAGRANGE's EOM

Total Kinetic Energy

8-2 EQUATIONS OF MOTION IN TERMS OF EULERIAN ANGLES

Now let us obtain expressions for ω_x, ω_y, and ω_z by summing the orthogonal projections of $\dot{\boldsymbol{\psi}}$, $\dot{\boldsymbol{\theta}}$, and $\dot{\boldsymbol{\phi}}$ onto each of the axes of the xyz system. We find that

$$\begin{aligned} \omega_x &= \dot{\phi} - \dot{\psi}\sin\theta \\ \omega_y &= \dot{\theta}\cos\phi + \dot{\psi}\cos\theta\sin\phi \\ \omega_z &= \dot{\psi}\cos\theta\cos\phi - \dot{\theta}\sin\phi \end{aligned} \tag{8–85}$$

$$\begin{aligned} T_{\text{rot}} = \frac{1}{2}I_{xx}(\dot{\phi} - \dot{\psi}\sin\theta)^2 + \frac{1}{2}I_{yy}(\dot{\theta}\cos\phi + \dot{\psi}\cos\theta\sin\phi)^2 \\ + \frac{1}{2}I_{zz}(\dot{\psi}\cos\theta\cos\phi - \dot{\theta}\sin\phi)^2 \end{aligned} \tag{8–87}$$

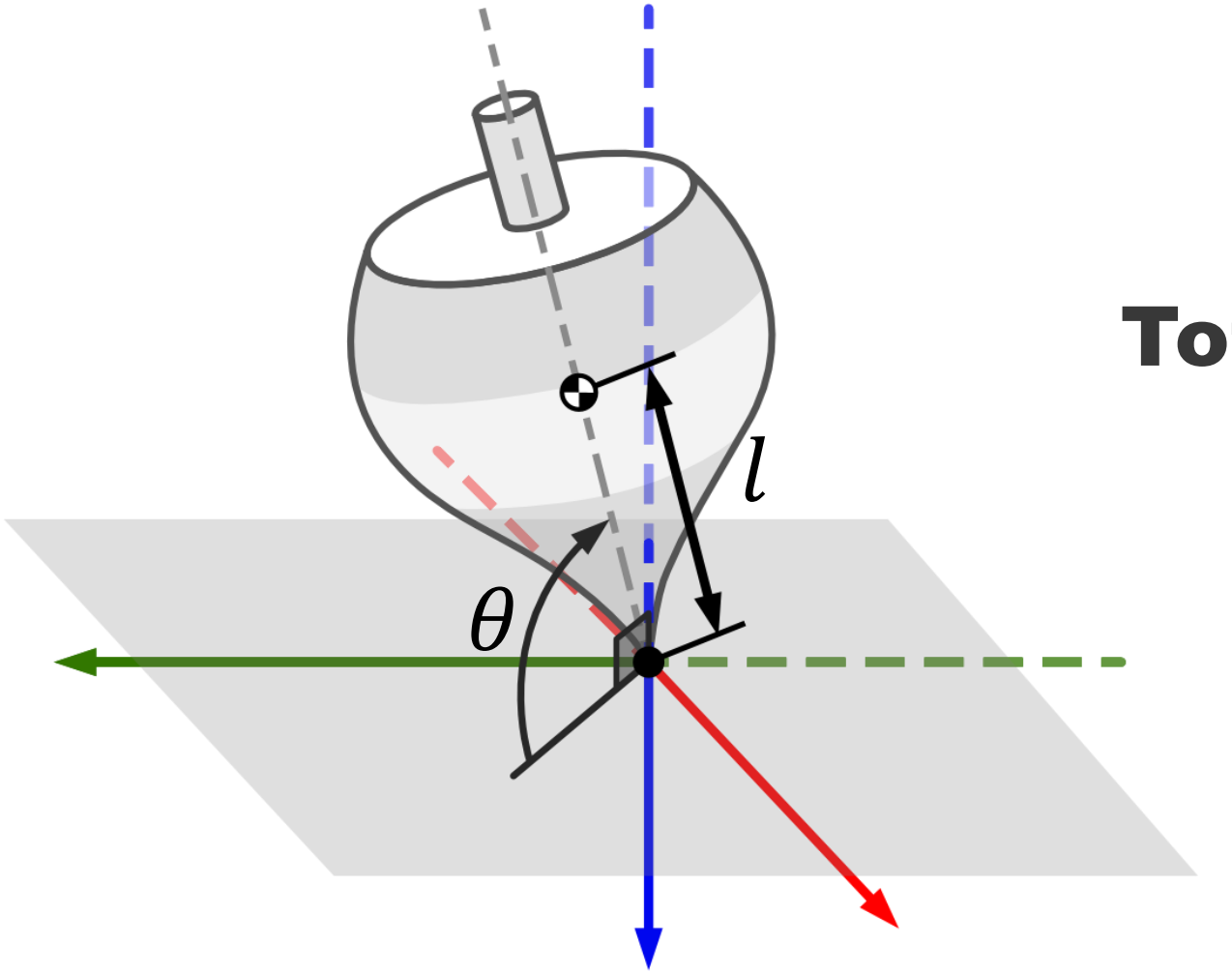
8-5 THE MOTION OF A TOP

$I_{xx} = I_a$ and $I_{yy} = I_{zz} = I_t$

symmetry about the x axis,

$$\begin{aligned} T = \frac{1}{2}I_a(\dot{\phi} - \dot{\psi}\sin\theta)^2 \\ + \frac{1}{2}I_t(\dot{\theta}^2 + \dot{\psi}^2\cos^2\theta) \end{aligned} \tag{8–140}$$

DERIVATION OF LAGRANGE's EOM



Total Potential Energy

$$V = mgl\sin\theta$$

DERIVATION OF LAGRANGE's EOM

Lagrangian Equation

$$L = T - V$$

$$L = \frac{1}{2}\left(I_a\left(\dot{\phi} - \dot{\psi}\sin\theta\right)^2 + I_t\left(\dot{\theta}^2 + \dot{\psi}^2\cos^2\theta\right)\right) - mgl\sin\theta$$

DERIVATION OF LAGRANGE's EOM

$$L = \frac{1}{2}\left(I_a\left(\dot{\phi} - \dot{\psi}\sin\theta\right)^2 + I_t\left(\dot{\theta}^2 + \dot{\psi}^2\cos^2\theta\right)\right) - mgl\sin\theta$$

Generalize Angular Momentum

$$p_\psi = \frac{\partial L}{\partial \dot{\psi}} = -I_a\Omega\, sin\,\theta + I_t\dot{\psi}\cos^2\theta = \text{Const.}$$

$$p_\phi = \frac{\partial L}{\partial \dot{\phi}} = I_a\left(\dot{\phi} - \dot{\psi}\sin\theta\right) = I_a\Omega = \text{Const.}$$

$$p_\theta = \frac{\partial L}{\partial \dot{\theta}} = I_t\dot{\theta}$$

Momentum Conserved

$$\Omega = \dot{\phi} - \dot{\psi}\sin\theta = \text{constant}$$

$$\boxed{\dot{\psi} = \frac{p_\psi + I_a\Omega\, sin\,\theta}{I_t\, cos^2\,\theta}}$$

$$\boxed{\dot{\phi} = \Omega + \dot{\psi}\sin\theta}$$

DERIVATION OF LAGRANGE's EOM

$$L = \frac{1}{2}\left(I_a\left(\dot{\phi} - \dot{\psi}\sin\theta\right)^2 + I_t\left(\dot{\theta}^2 + \dot{\psi}^2\cos^2\theta\right)\right) - mgl\sin\theta \quad \text{And} \quad p_\theta = \frac{\partial L}{\partial \dot{\theta}} = I_t\dot{\theta}$$

Euler-Lagrange Equation

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \left(\frac{\partial L}{\partial \theta}\right) = 0$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) = \frac{d}{dt}(p_\theta) = I_t\ddot{\theta} \quad \Big| \quad \frac{\partial L}{\partial \theta} = -I_a\left(\dot{\phi} - \dot{\psi}\sin\theta\right)\dot{\psi}\cos\theta - I_t\dot{\psi}^2\sin\theta\cos\theta - mgl\cos\theta$$

$$I_t\ddot{\theta} + I_a\left(\dot{\phi} - \dot{\psi}\sin\theta\right)\dot{\psi}\cos\theta + I_t\dot{\psi}^2\sin\theta\cos\theta + mgl\cos\theta = 0$$

$$\boxed{\ddot{\theta} = \frac{-I_a\left(\dot{\phi} - \dot{\psi}\sin\theta\right)\dot{\psi}\cos\theta - I_t\dot{\psi}^2\sin\theta\cos\theta - mgl\cos\theta}{I_t}}$$

NUMERICAL SIMULATION SETUP

State vector:

$$\boldsymbol{S} = [S_1, S_2, S_3, S_4]^T = \left[\theta, \psi, \phi, \dot{\theta}\right]^T$$

$$\dot{\boldsymbol{S}} = \left[\dot{S}_1, \dot{S}_2, \dot{S}_3, \dot{S}_4\right]^T$$

where:

$$\dot{S}_1 = \dot{\theta} = S_4$$

$$\dot{S}_2 = \dot{\psi} = \frac{p_\psi + I_a \Omega \sin\theta}{I_t \cos^2\theta}$$

$$\dot{S}_3 = \dot{\phi} = \Omega + \dot{\psi} \sin\theta$$

$$\dot{S}_4 = \ddot{\theta} = \frac{-I_a\left(\dot{\phi} - \dot{\psi}\sin\theta\right)\dot{\psi}\cos\theta - I_t\dot{\psi}^2 \sin\theta\cos\theta - mgl\cos\theta}{I_t}$$

Table 1 The physical parameters used in simulation

Physical Parameters	Symbol	Value	Units
Mass of the top	m	1.0	kg
Distance between point O to point C	l	1.0	m
Gravitational acceleration	g	9.81	$\mathrm{m/s^2}$
Axial Moment of Inertia	I_a	1.0	$\mathrm{kg.m^2}$
Transverse Moment of Inertia	I_t	5.0	$\mathrm{kg.m^2}$

SIMULATION CASE

Define

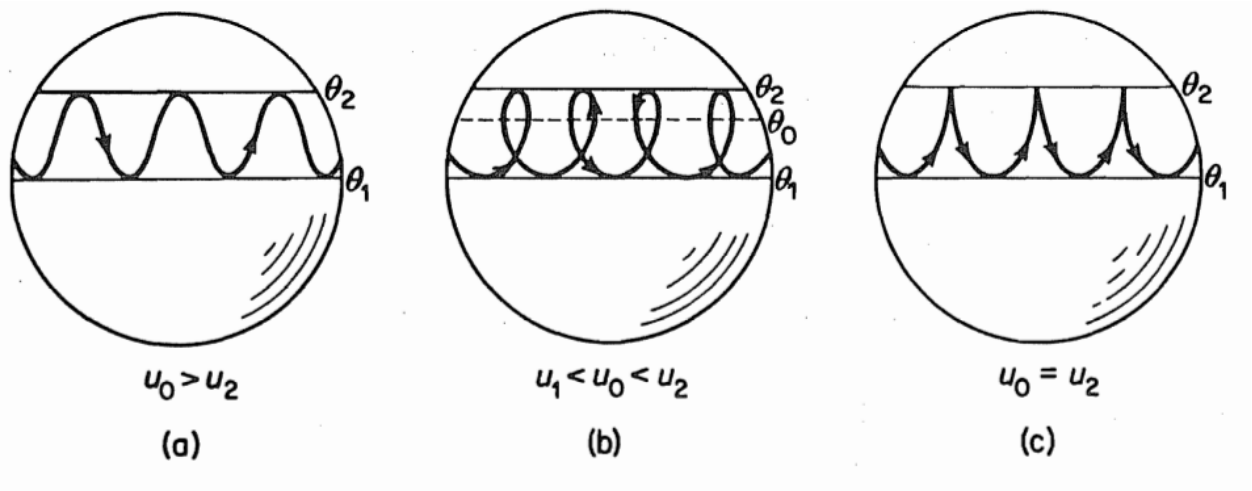
$u = sin\,\theta$ represent the vertical height of the top's tip

u_1 and u_2 is physical boundaries (the lowest and highest points)

u_0 is the exact height where the top stops sweeping forward

$$u_0 = -\frac{p_\psi}{p_\phi}$$

Possible Path of the Axis of Symmetry (Fig. 8-15)



CASE #1

Generating Paths as Shown in Figure 8-15

CASE #2

Both Case of Cuspidal Motion

CASE #3

Stability of motion near the vertical

CASE #4

Nutation and Amplitude for the case of large spin

CASE #5

Precession with no Nutation

SIMULATION CASE#1

Generating Paths as Shown in Figure 8-15

Set: $\dot{\theta} = 0$

Case (a) $u_0 > u_2$

$$\theta(0) = 60°, \qquad \dot{\psi}(0) = -1.0 \ \ \mathrm{rad/s}, \qquad \Omega = 12.0 \ \ \mathrm{rad/s}$$

Case (b) $u_1 < u_0 < u_2$

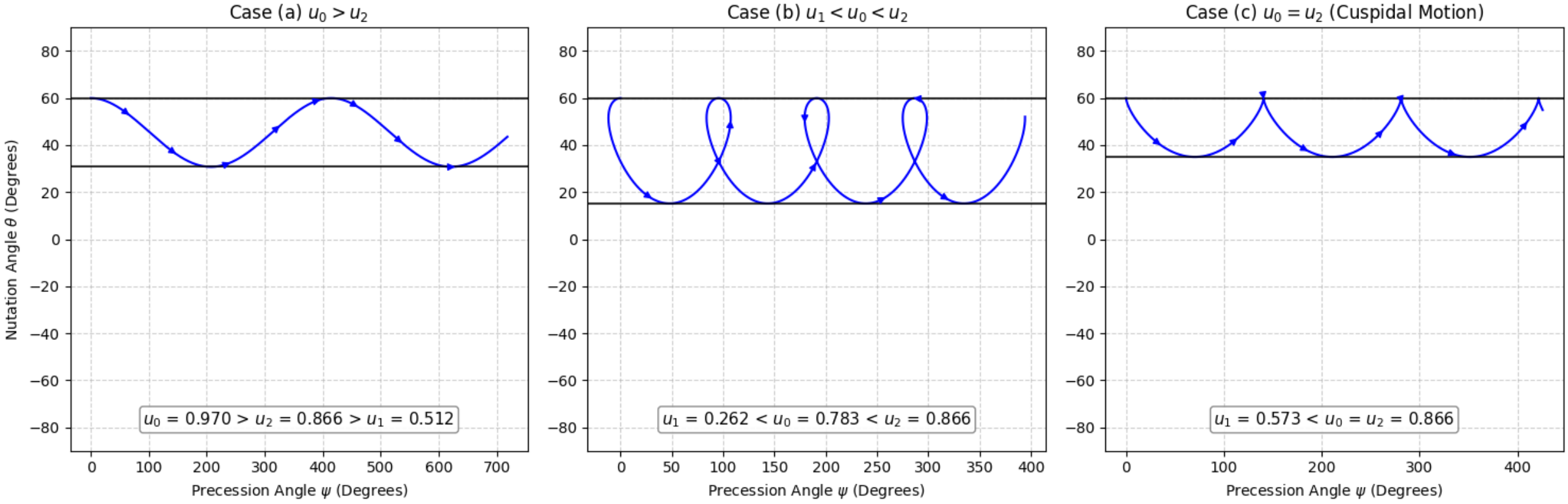
$$\theta(0) = 60°, \qquad \dot{\psi}(0) = 1.0\,\mathrm{rad/s}, \qquad \Omega = 15.0\,\mathrm{rad/s}$$

Case (c) $u_0 = u_2$

$$\theta(0) = 60°, \qquad \dot{\psi}(0) = 0.0\,\mathrm{rad/s}, \qquad \Omega = 15.0\,\mathrm{rad/s}$$

SIMULATION CASE#1

Generating Paths as Shown in Figure 8-15



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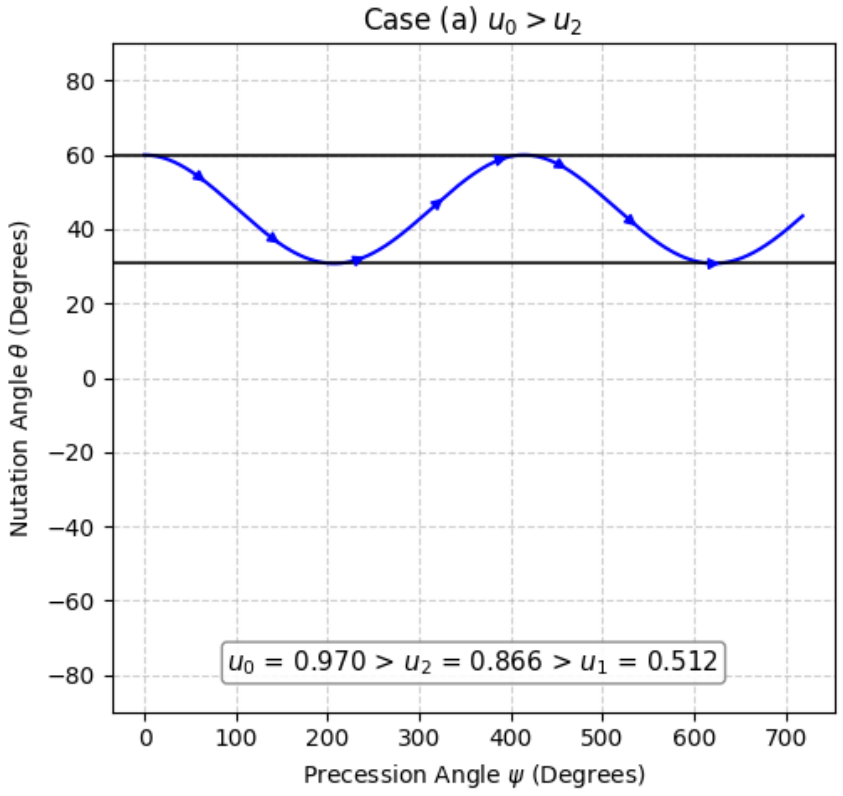
SIMULATION CASE#1

Generating Paths as Shown in Figure 8-15



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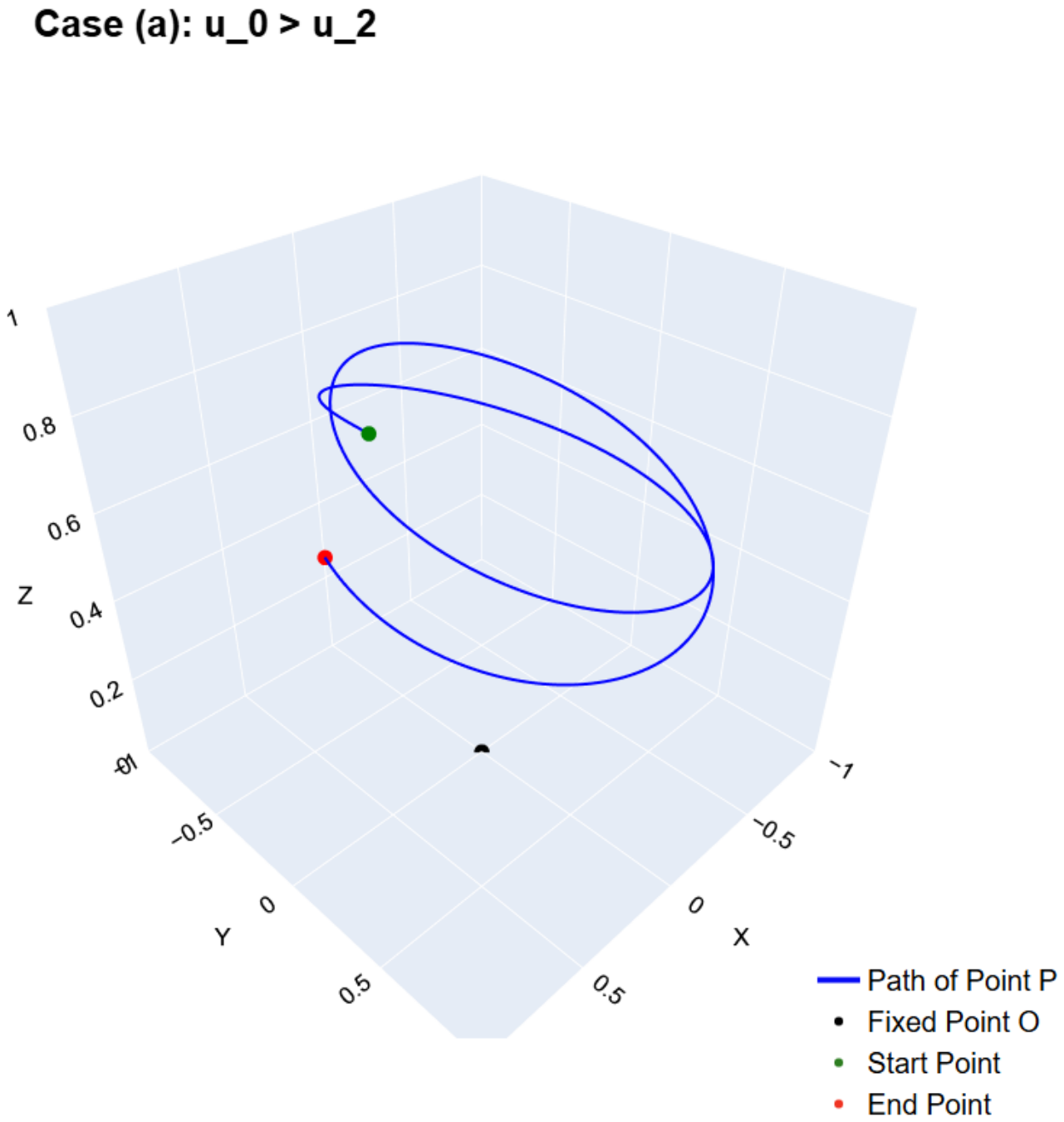
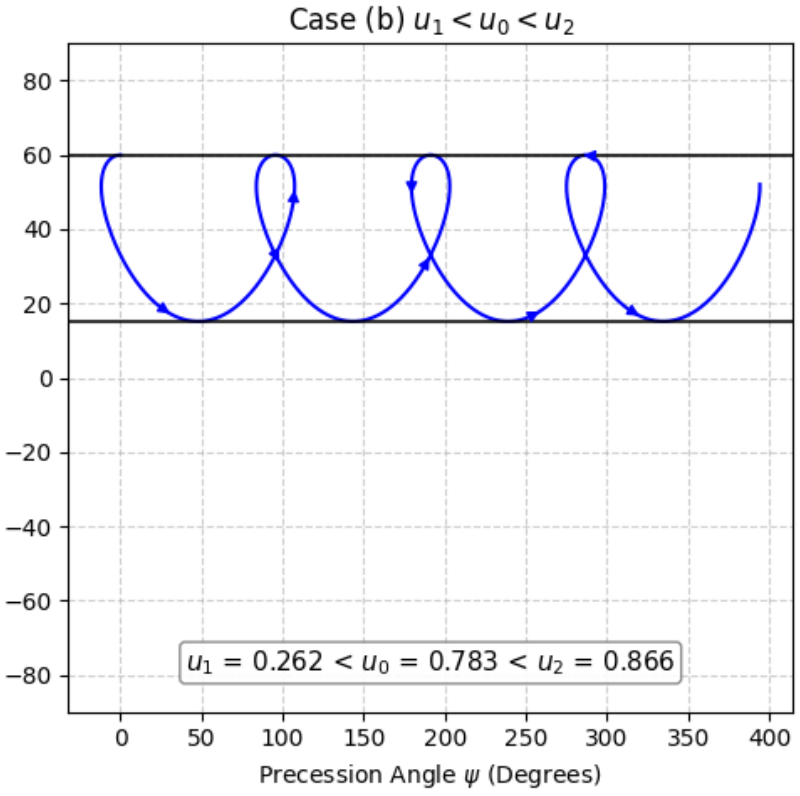


SIMULATION CASE#1

Generating Paths as Shown in Figure 8-15



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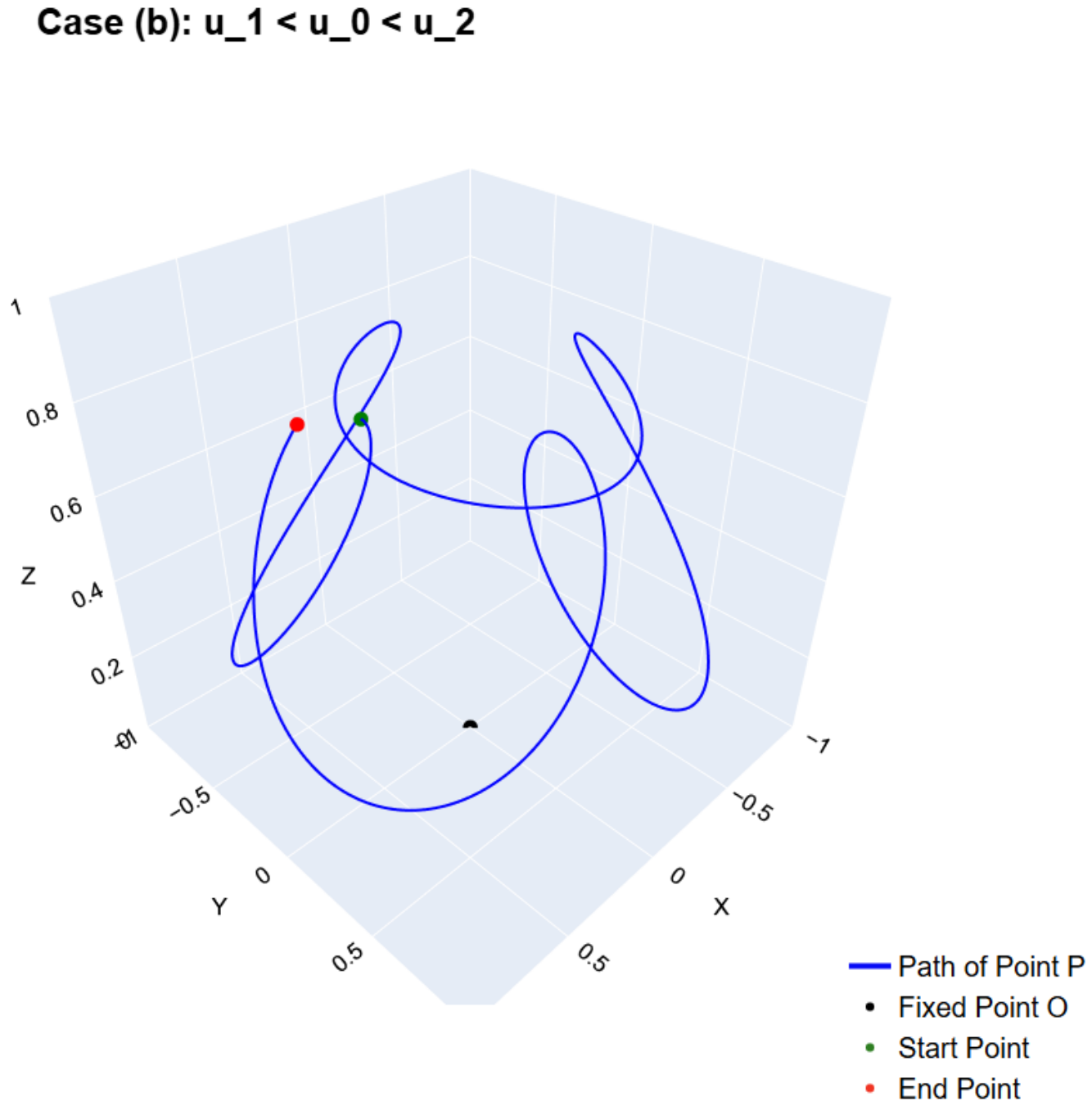
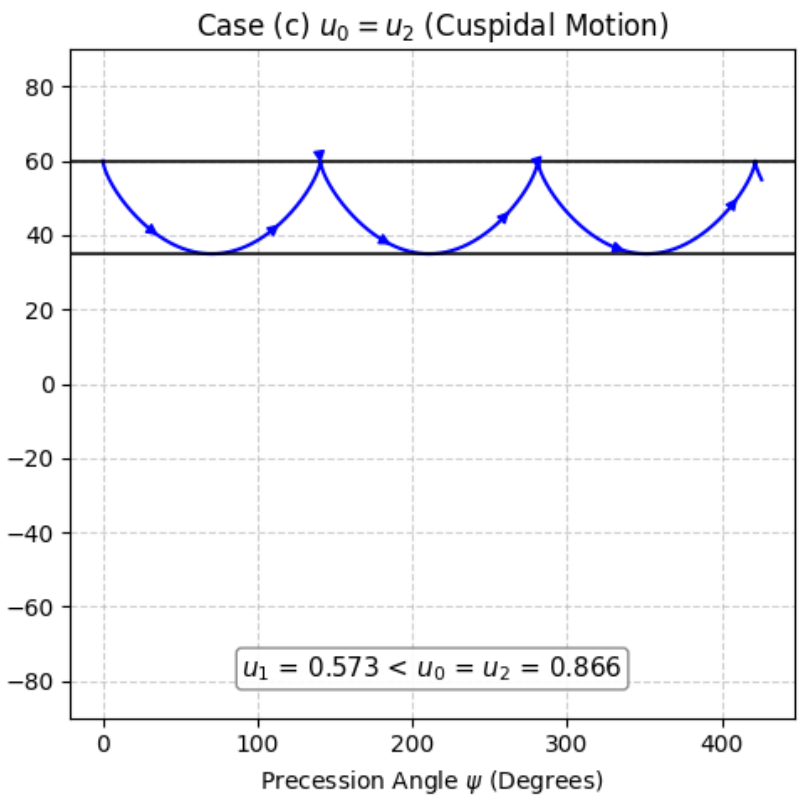
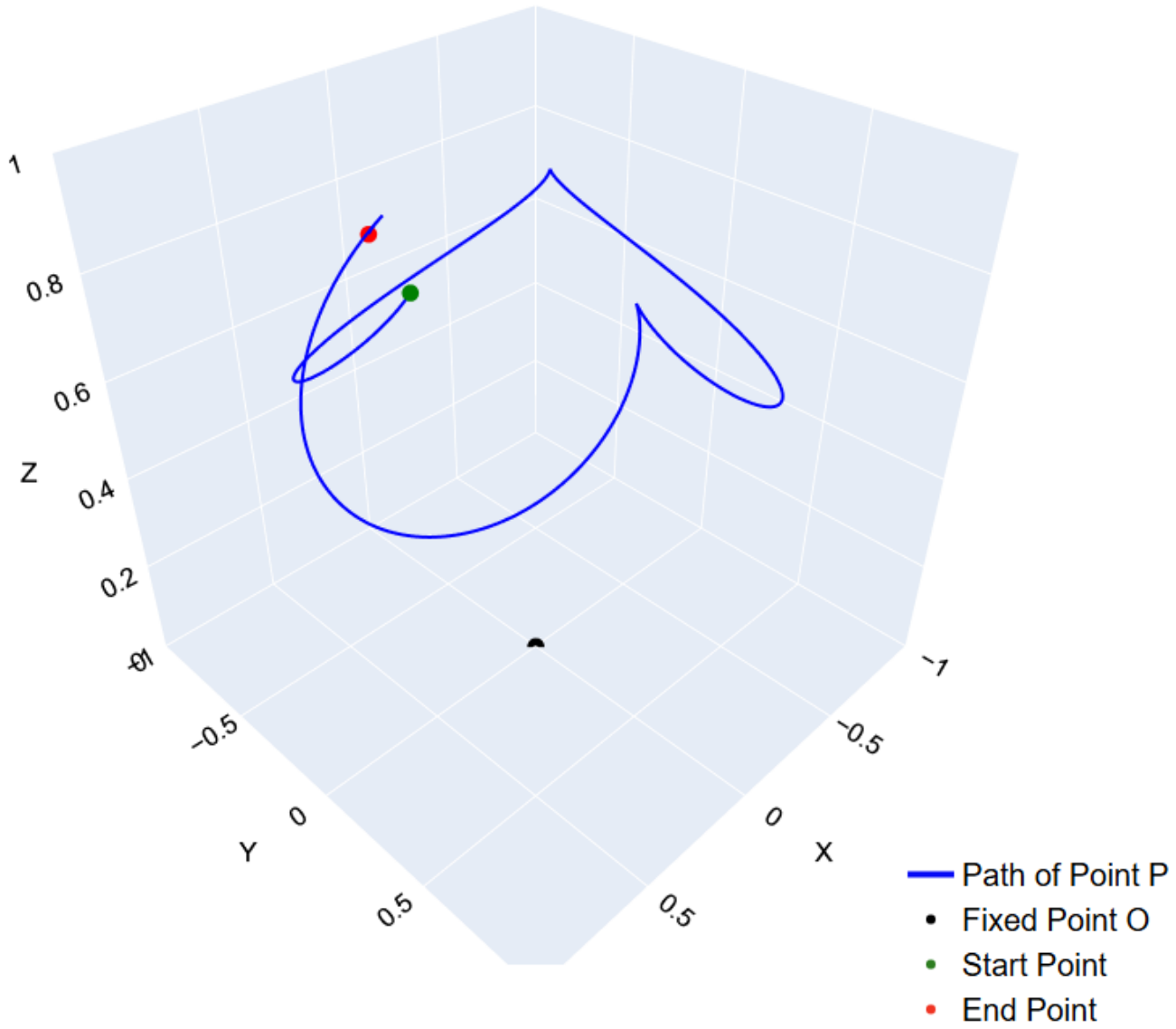


SIMULATION CASE#1

Generating Paths as Shown in Figure 8-15



$$\theta(0) = 60°$$
$$\dot{\psi}(0) = 0 \ \ \text{rad/s}$$
$$\Omega = 15.0 \ \ \text{rad/s}$$



SIMULATION CASE#2

Both Case of Cuspidal Motion

Case 1: $u = u_0$ when $\dot{\theta} = 0$

$$\theta(0) = 45°$$

$$\dot{\theta}(0) = 0 \text{ rad/s}$$

$$\dot{\psi}(0) = 0 \text{ rad/s}$$

$$\Omega = 20.0 \text{ rad/s}$$

Case 2: $u = u_0 = 1$ when $\dot{\theta} \neq 0$

$$\theta(0) = 90°$$

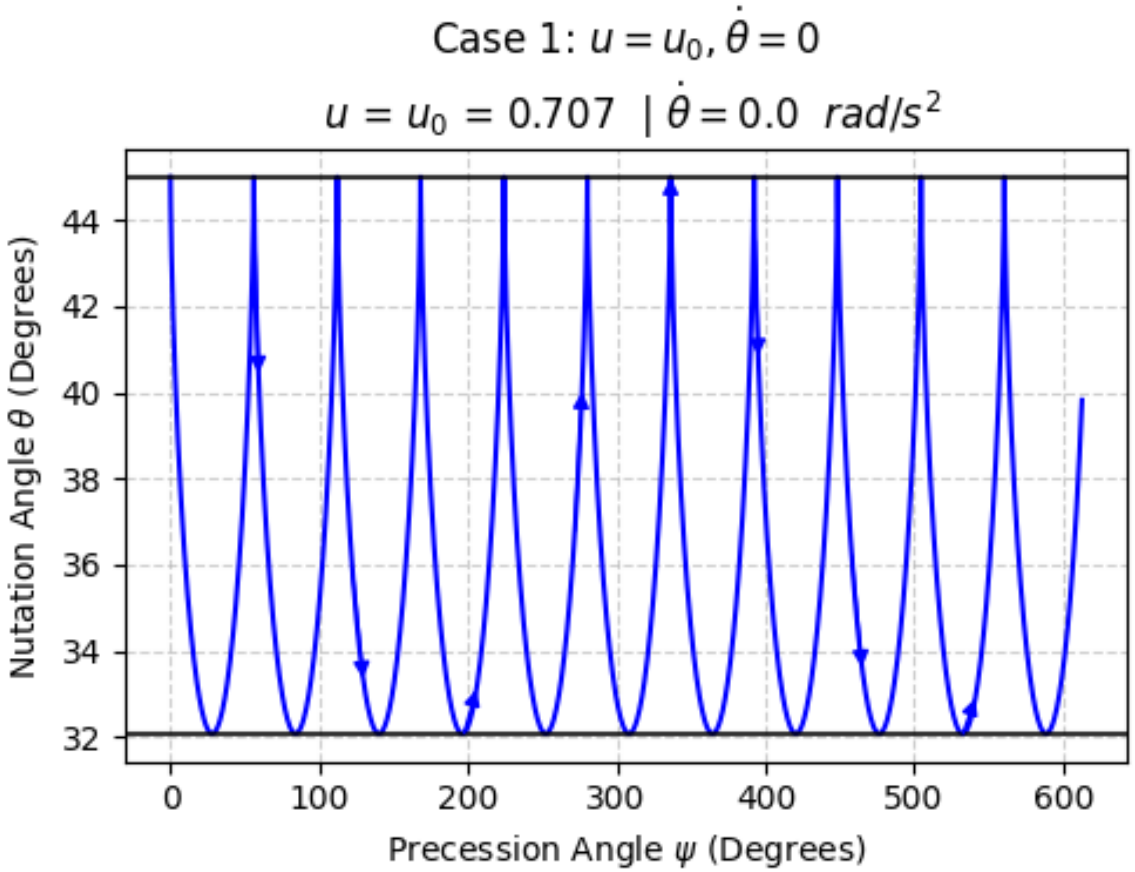
$$\dot{\theta}(0) = 1.0 \text{ rad/s}$$

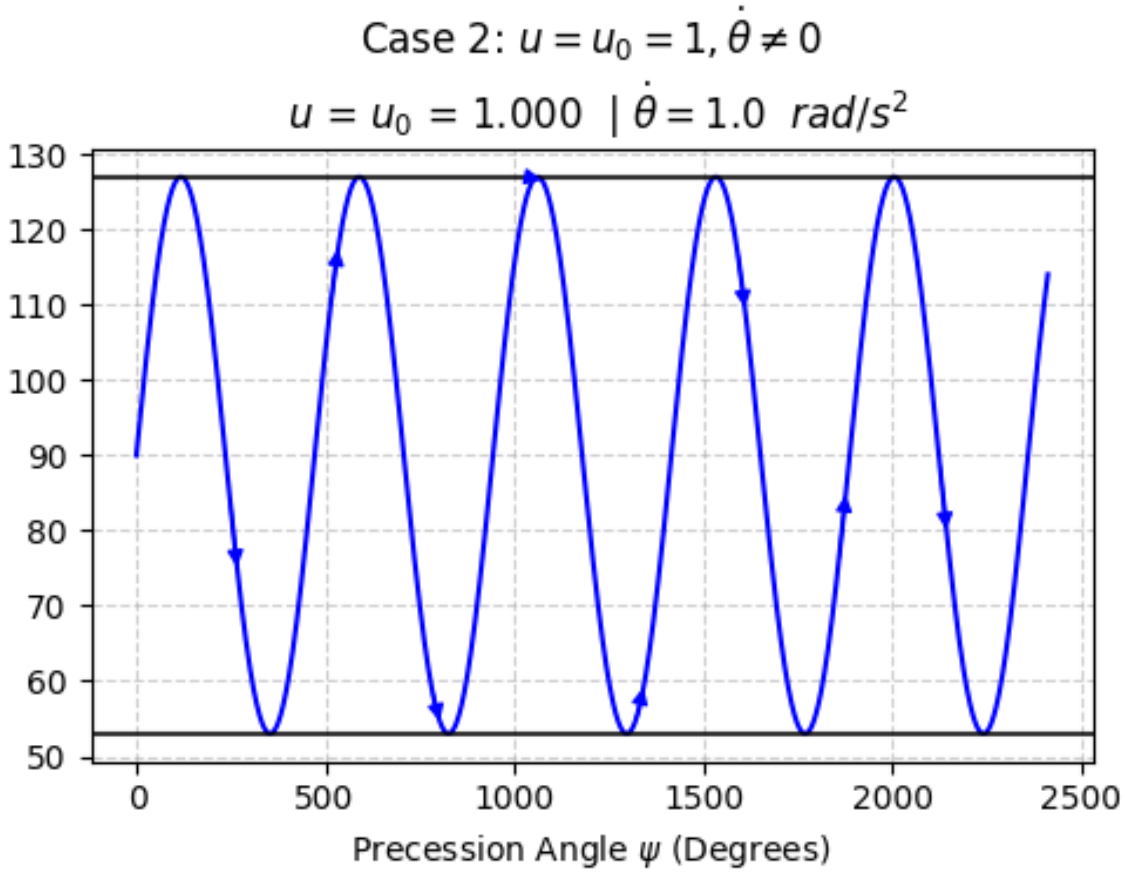
$$\dot{\psi}(0) = 0 \text{ rad/s}$$

$$\Omega = 20.0 \text{ rad/s}$$

SIMULATION CASE#2

Both Case of Cuspidal Motion





$\theta(0) = 45°, \qquad \dot{\theta}(0) = 0 \text{ rad/s}$

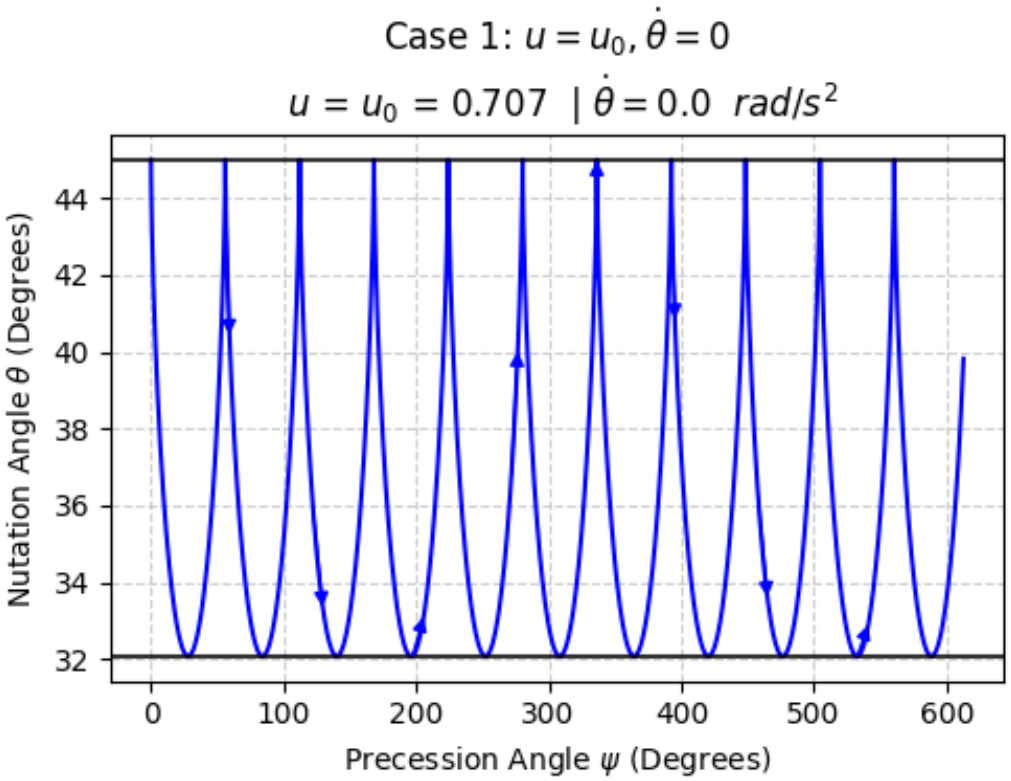
$\dot{\psi}(0) = 0 \text{ rad/s}, \qquad \Omega = 20.0 \text{ rad/s}$

$\theta(0) = 90°, \qquad \dot{\theta}(0) = 1.0 \text{ rad/s}$

$\dot{\psi}(0) = 0 \text{ rad/s}, \qquad \Omega = 20.0 \text{ rad/s}$

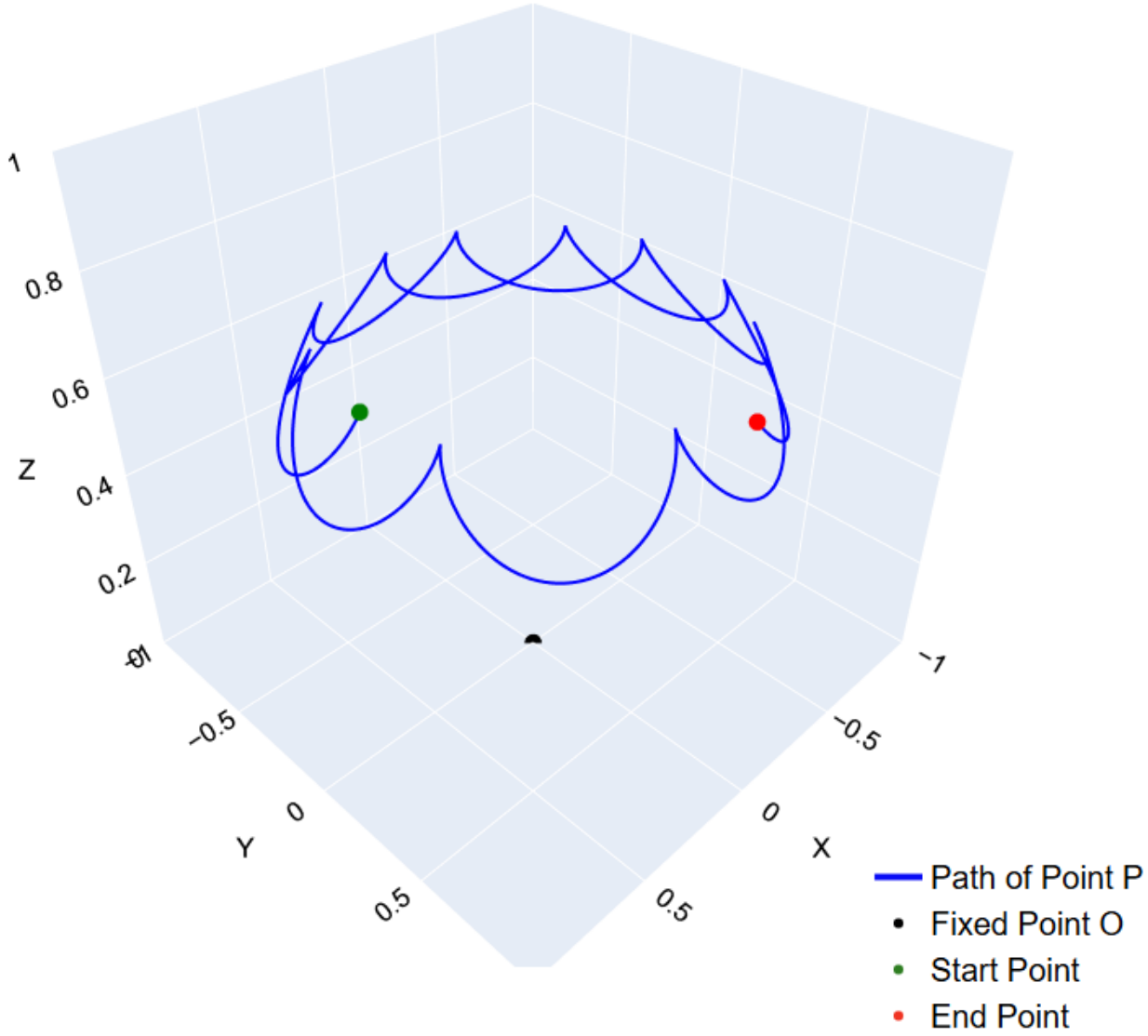
SIMULATION CASE#2

Both Case of Cuspidal Motion



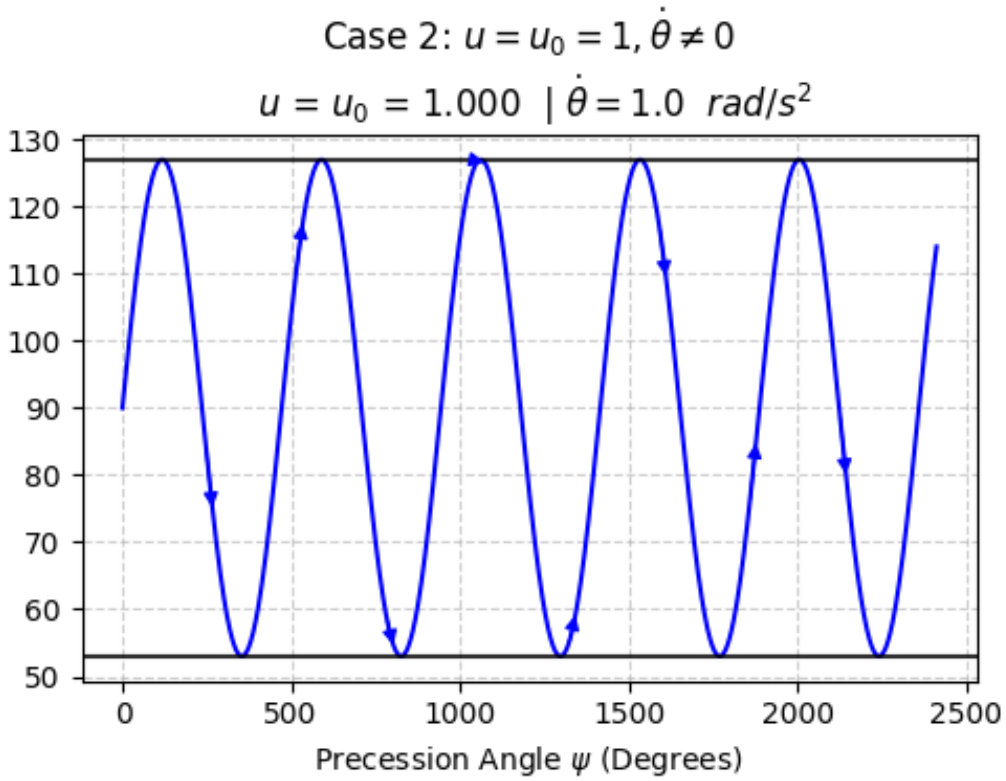
$$\theta(0) = 45°, \qquad \dot{\theta}(0) = 0 \text{ rad/s}$$

$$\dot{\psi}(0) = 0 \text{ rad/s}, \qquad \Omega = 20.0 \text{ rad/s}$$



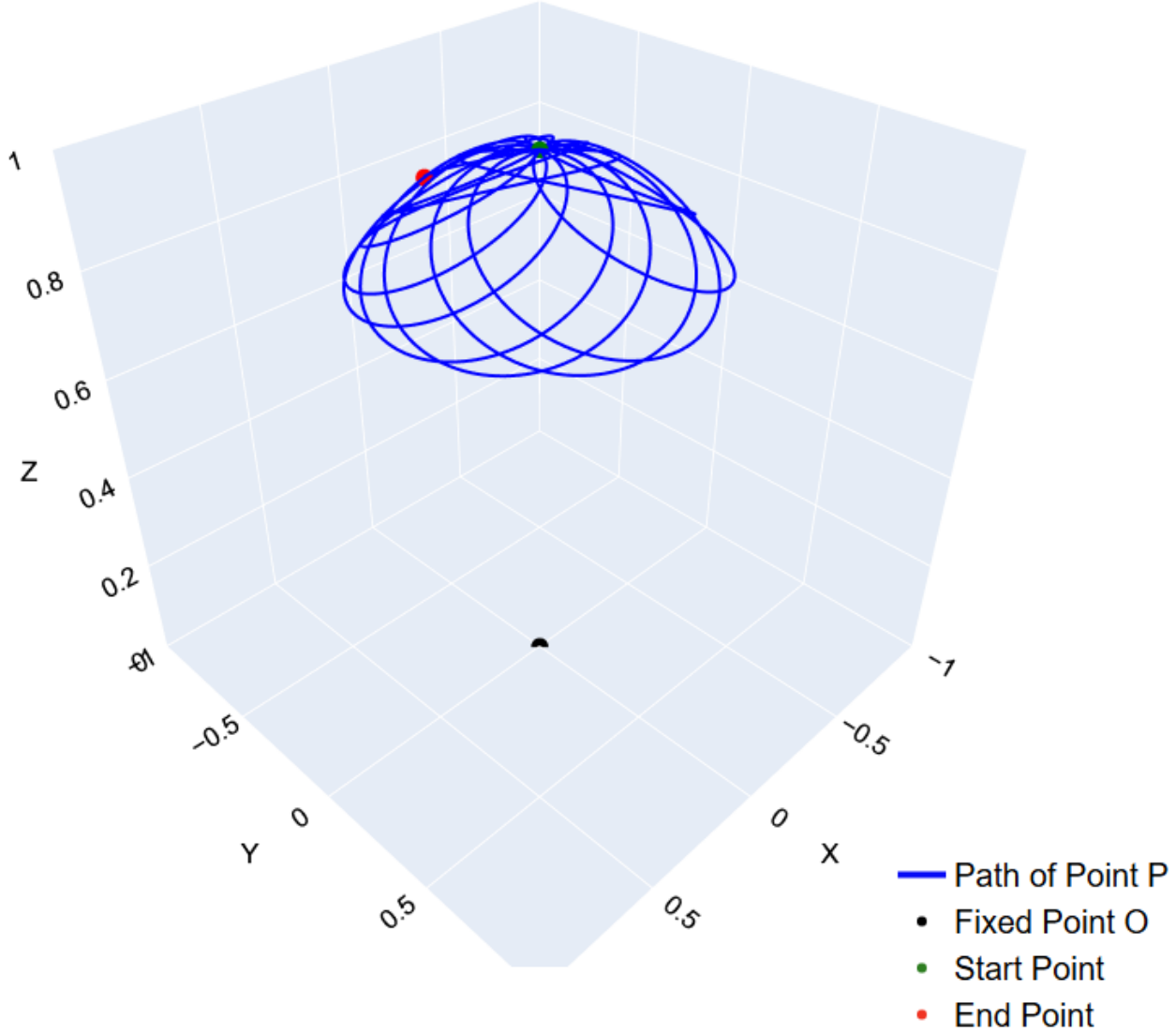
SIMULATION CASE#2

Both Case of Cuspidal Motion



$$\theta(0) = 90°, \qquad \dot{\theta}(0) = 1.0 \text{ rad/s}$$

$$\dot{\psi}(0) = 0 \text{ rad/s}, \qquad \Omega = 20.0 \text{ rad/s}$$



SIMULATION CASE#3

Stability of motion near the vertical

$$\lambda = \frac{I_a^2 \Omega^2}{4 I_t m g l}$$

$$\text{Let } \Omega_{\text{critical}} = \sqrt{4 I_t m g l / I_a^2} \text{ be critical spin rate}$$

Case 1: $\lambda < 1$ or $\Omega^2 < 4 I_t m g l / I_a^2$ or $\Omega < \Omega_{\text{critical}}$

$\theta(0) = 85°$, $\dot{\theta}(0) = 0\,\text{rad/s}$, $\dot{\psi}(0) = 0\ \text{rad/s}$, $\Omega = 0.8\,\Omega_{\text{critical}}$

Case 2: $\lambda = 1$ or $\Omega^2 = 4 I_t m g l / I_a^2$ or $\Omega = \Omega_{\text{critical}}$

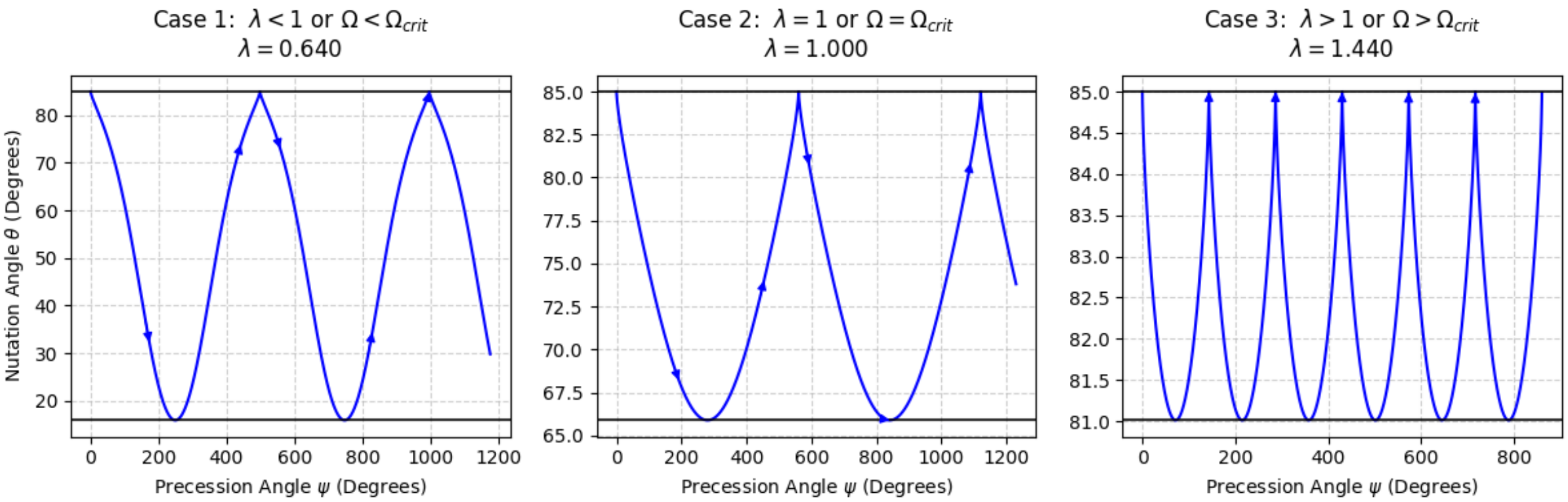
$\theta(0) = 85°$, $\dot{\theta}(0) = 0\,\text{rad/s}$, $\dot{\psi}(0) = 0\ \text{rad/s}$, $\Omega = \Omega_{\text{critical}}$

Case 3: $\lambda > 1$ or $\Omega^2 > 4 I_t m g l / I_a^2$ or $\Omega > \Omega_{\text{critical}}$

$\theta(0) = 85°$, $\dot{\theta}(0) = 0\,\text{rad/s}$, $\dot{\psi}(0) = 0\ \text{rad/s}$, $\Omega = 1.2\,\Omega_{\text{critical}}$

SIMULATION CASE#3

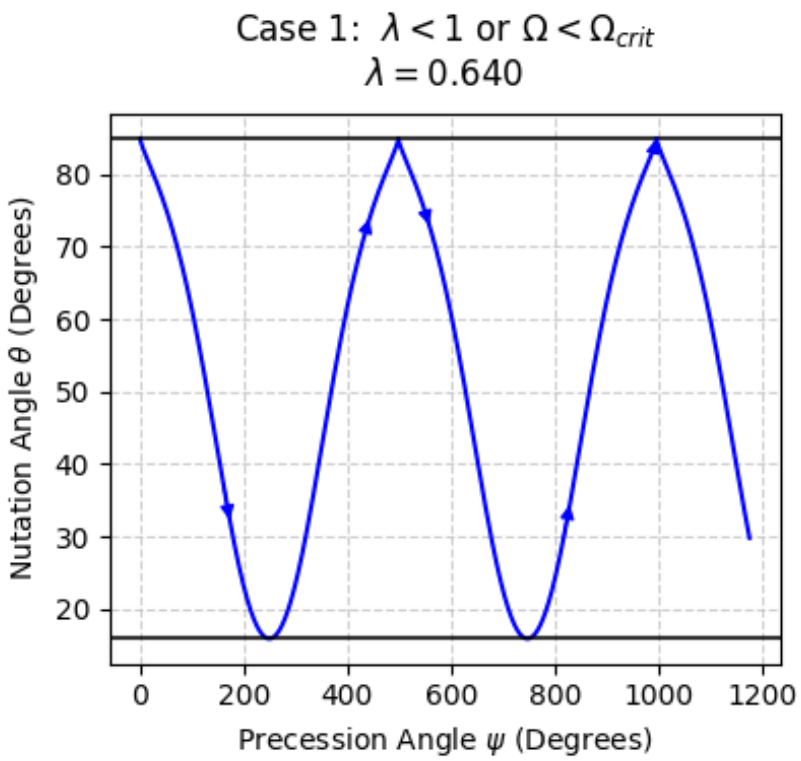
Stability of motion near the vertical



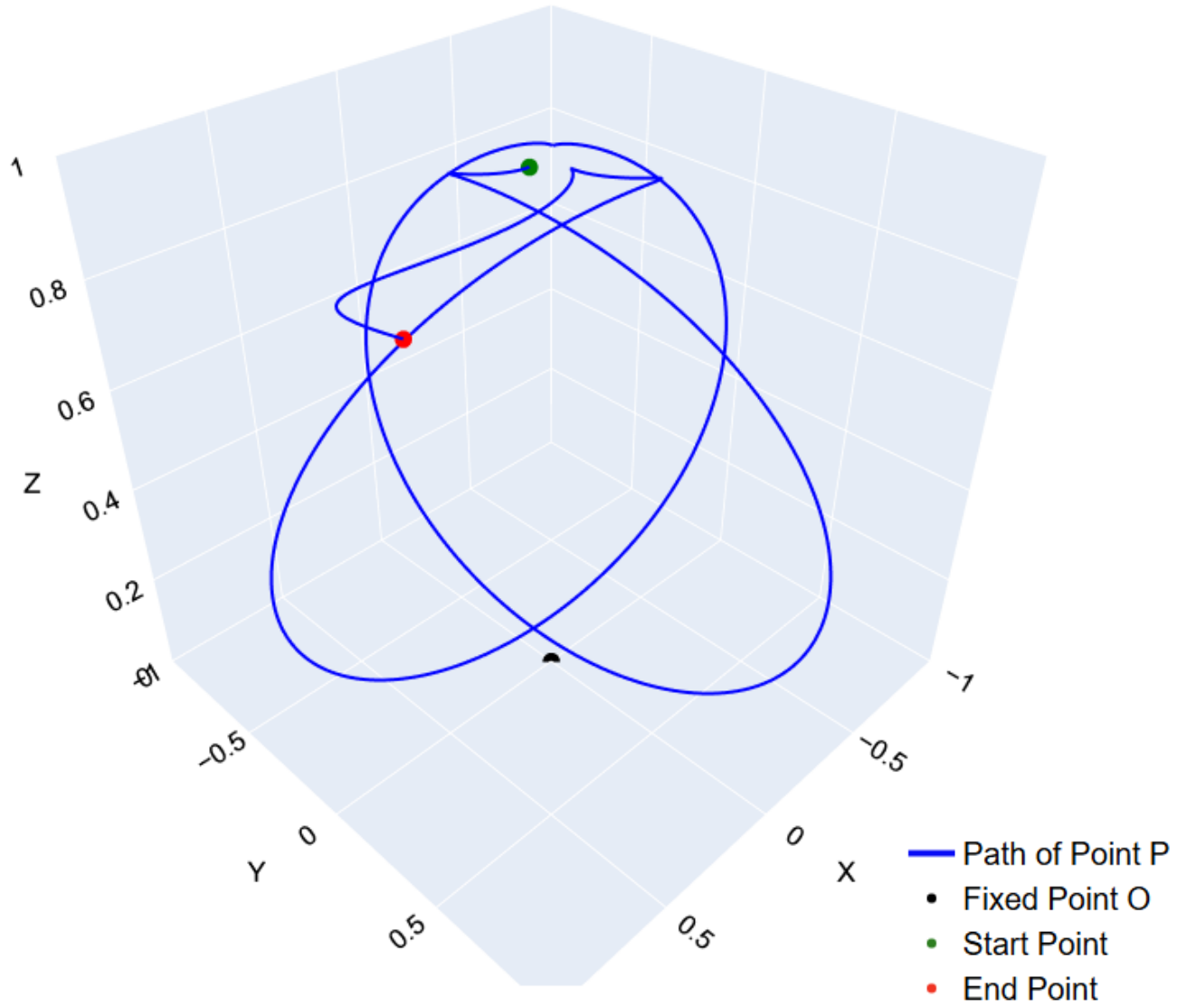
$\Omega = 0.8\,\Omega_{\text{critical}}$ $\Omega = \Omega_{\text{critical}}$ $\Omega = 1.2\,\Omega_{\text{critical}}$

SIMULATION CASE#3

Stability of motion near the vertical

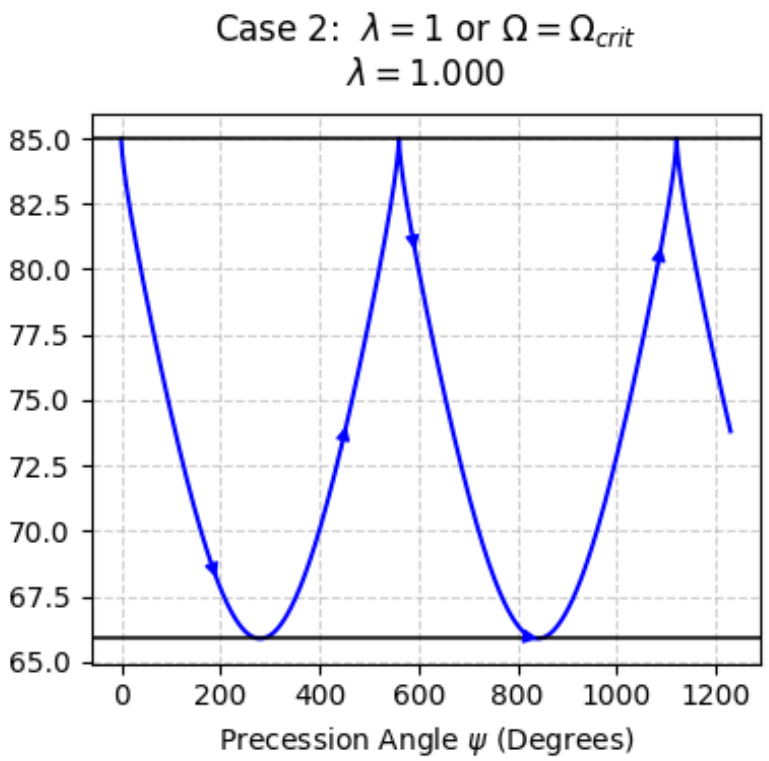


$$\Omega = 0.8\,\Omega_{\text{critical}}$$

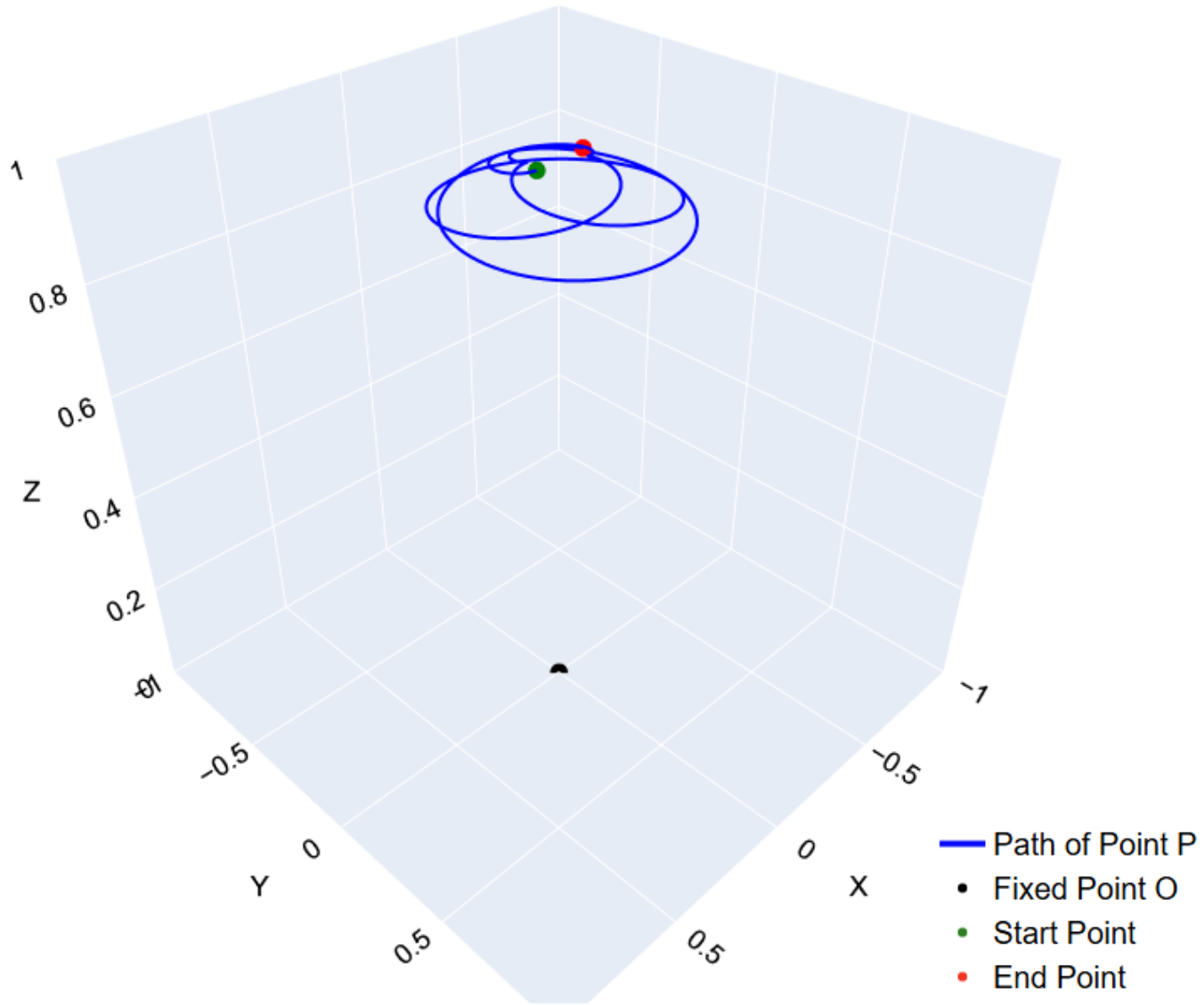


SIMULATION CASE#3

Stability of motion near the vertical

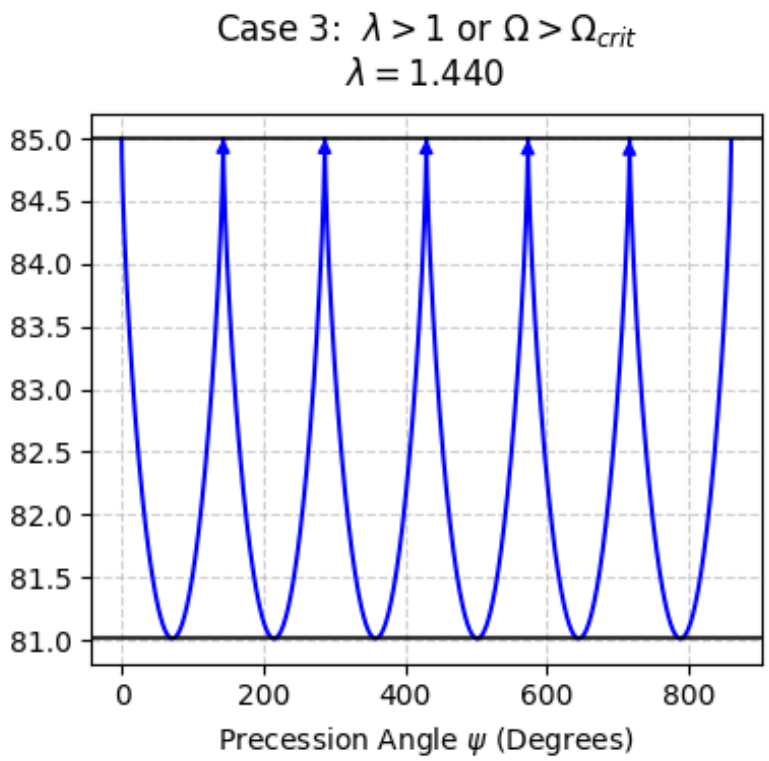


$$\Omega = \Omega_{\text{critical}}$$

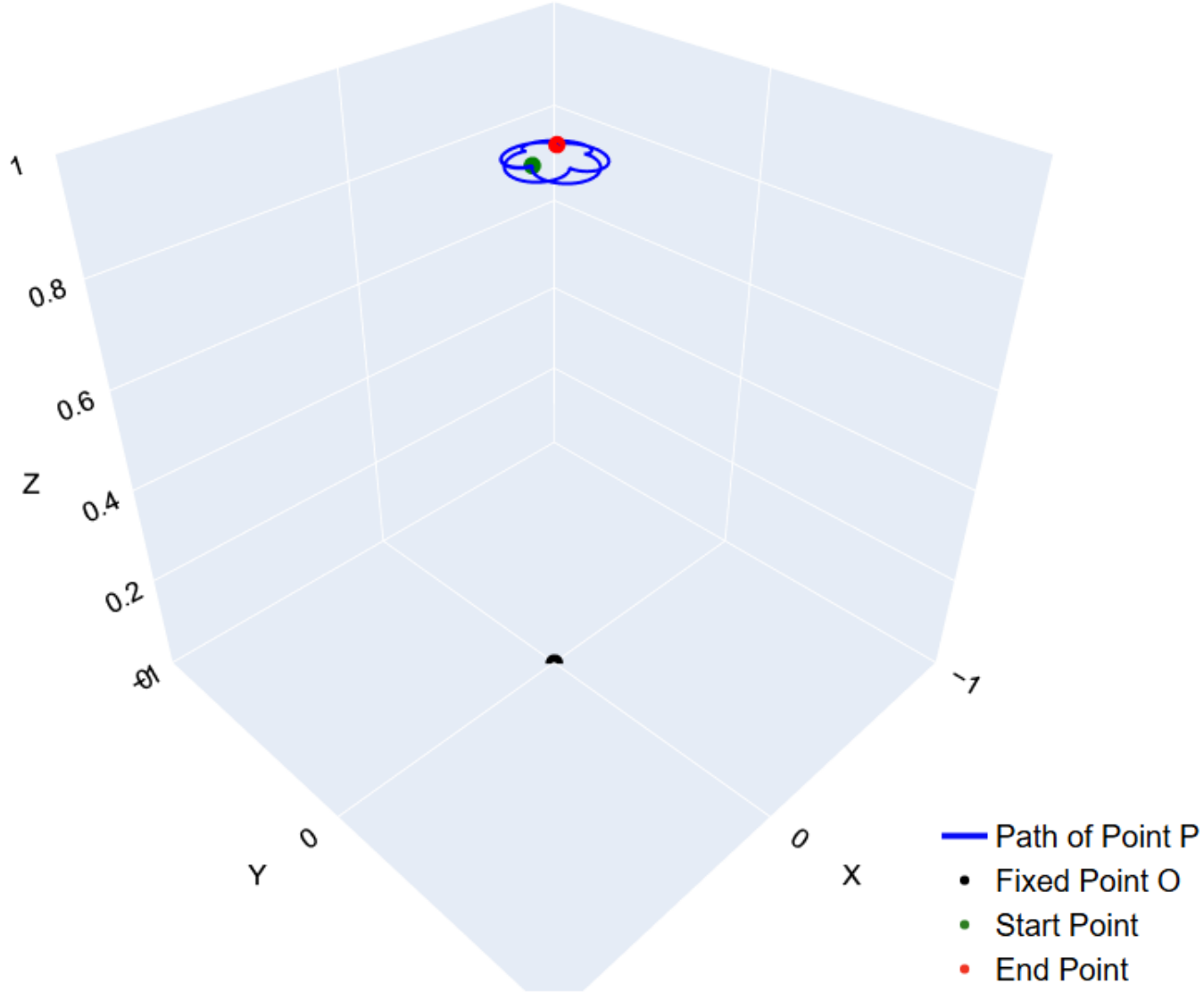


SIMULATION CASE#3

Stability of motion near the vertical



$$\Omega = 1.2\ \Omega_{\text{critical}}$$



SIMULATION CASE#4

Nutation and Amplitude for the case of large spin

The amplitude of the nutation: $u_0 - u_1 \cong \frac{1}{2\lambda}(1 - u_0^2) = \frac{2I_t mgl}{I_a^2 \Omega^2}(1 - u_0^2)$

The circular frequency of the nutation: $2p \cong \frac{I_a \Omega}{I_t}$

Case 1: Moderate Spin

$$\theta(0) = 60°, \qquad \dot{\theta}(0) = 0\,\mathrm{rad/s}, \qquad \dot{\psi}(0) = 0\ \mathrm{rad/s}, \qquad \Omega = 25\ \mathrm{rad/s}$$

Case 2: Large Spin

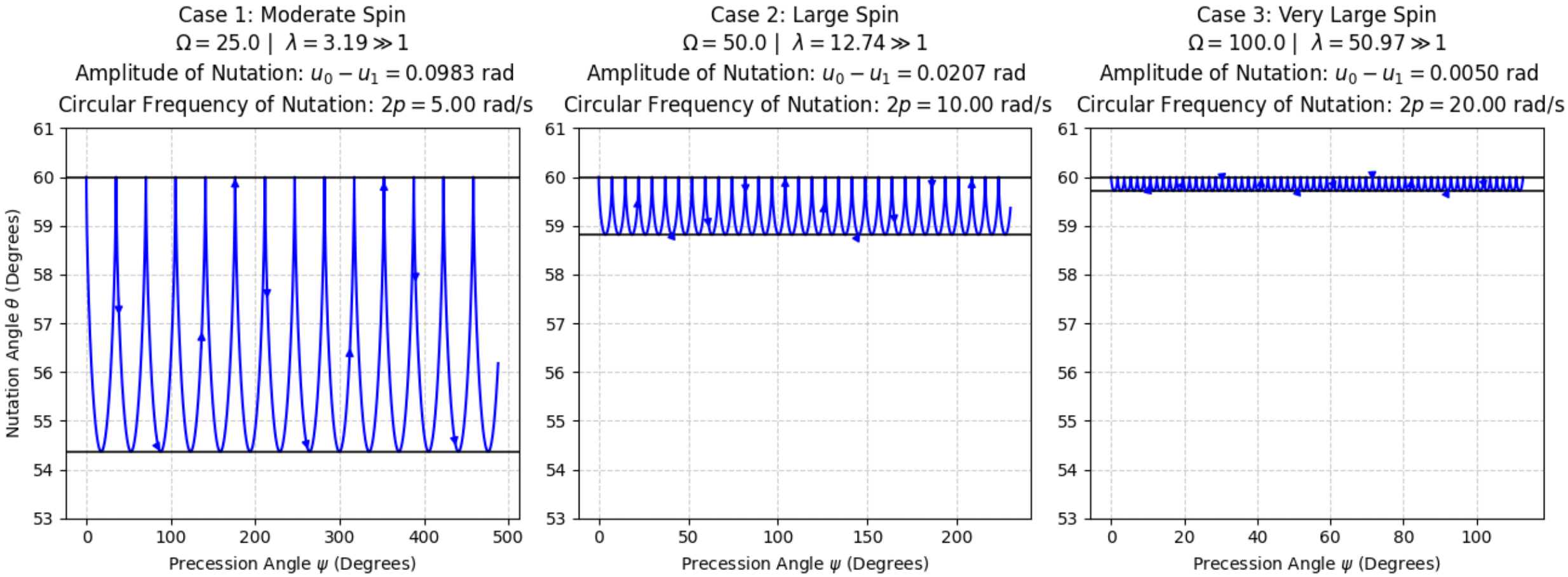
$$\theta(0) = 60°, \qquad \dot{\theta}(0) = 0\,\mathrm{rad/s}, \qquad \dot{\psi}(0) = 0\ \mathrm{rad/s}, \qquad \Omega = 50\ \mathrm{rad/s}$$

Case 3: Very Large Spin

$$\theta(0) = 60°, \qquad \dot{\theta}(0) = 0\,\mathrm{rad/s}, \qquad \dot{\psi}(0) = 0\ \mathrm{rad/s}, \qquad \Omega = 100\ \mathrm{rad/s}$$

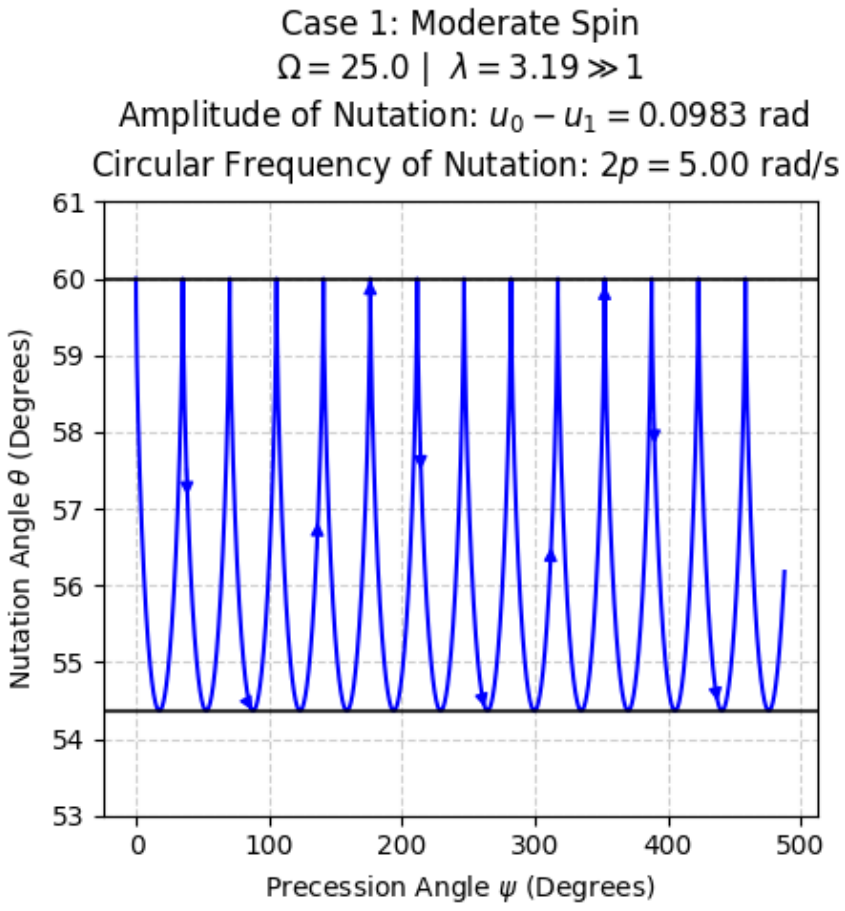
SIMULATION CASE#4

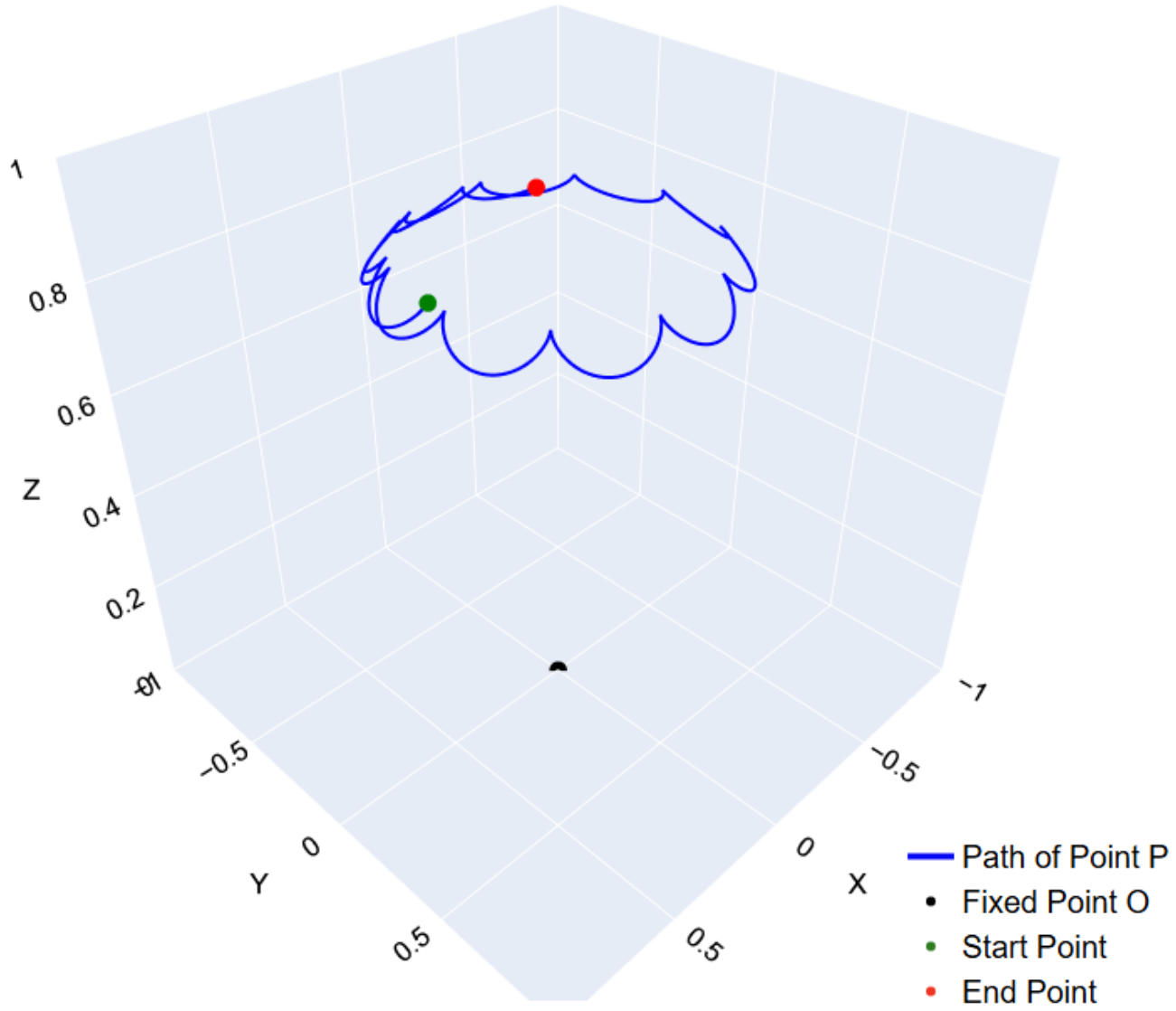
Nutation and Amplitude for the case of large spin



SIMULATION CASE#4

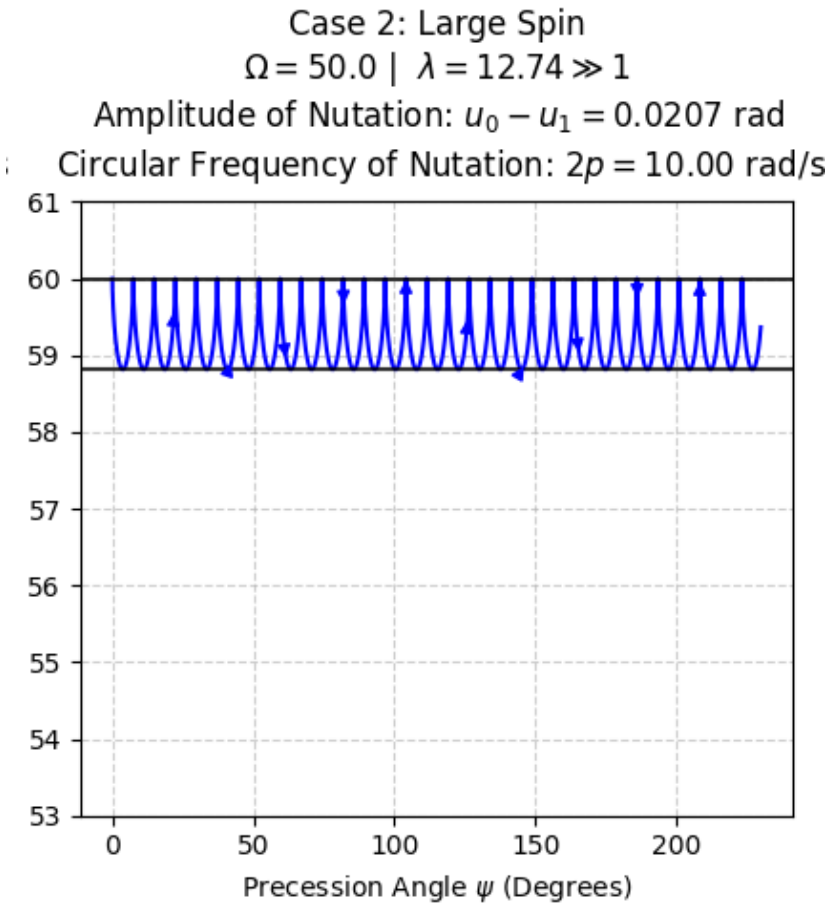
Nutation and Amplitude for the case of large spin

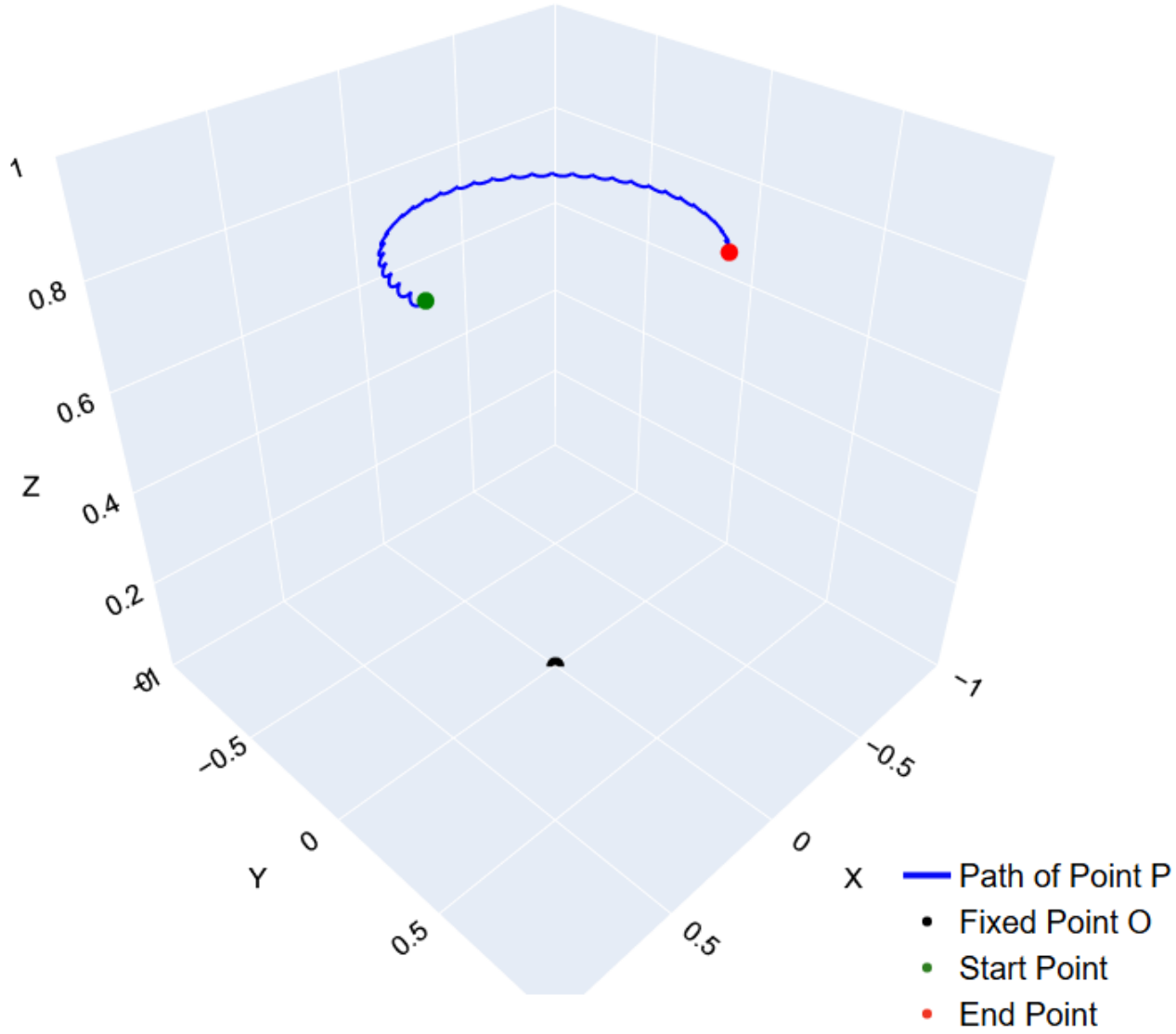




SIMULATION CASE#4

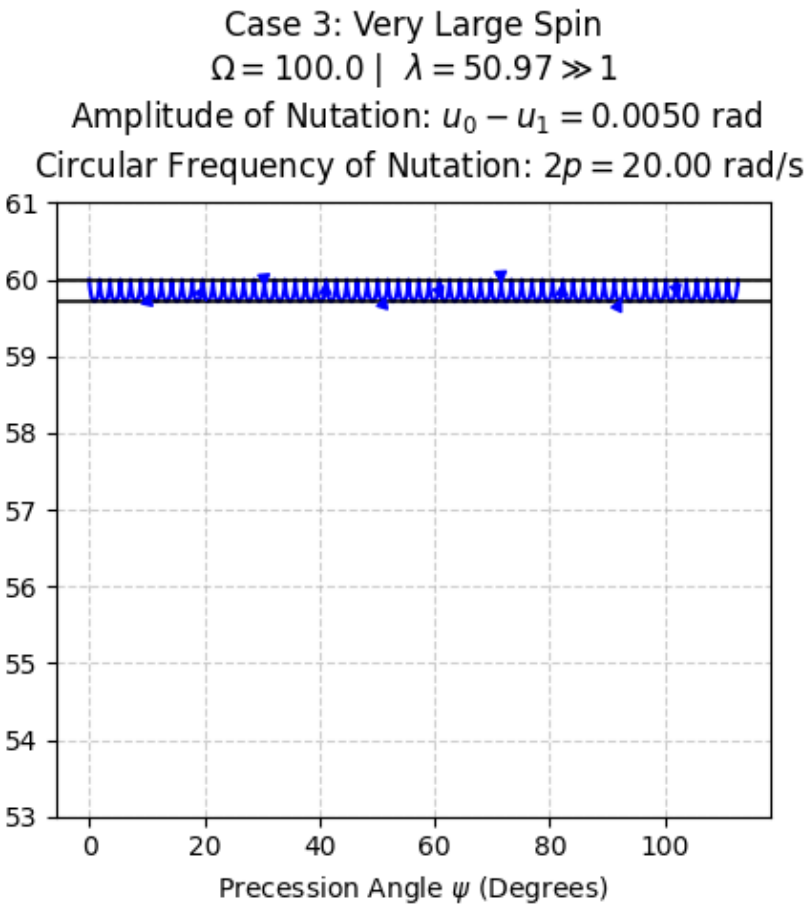
Nutation and Amplitude for the case of large spin

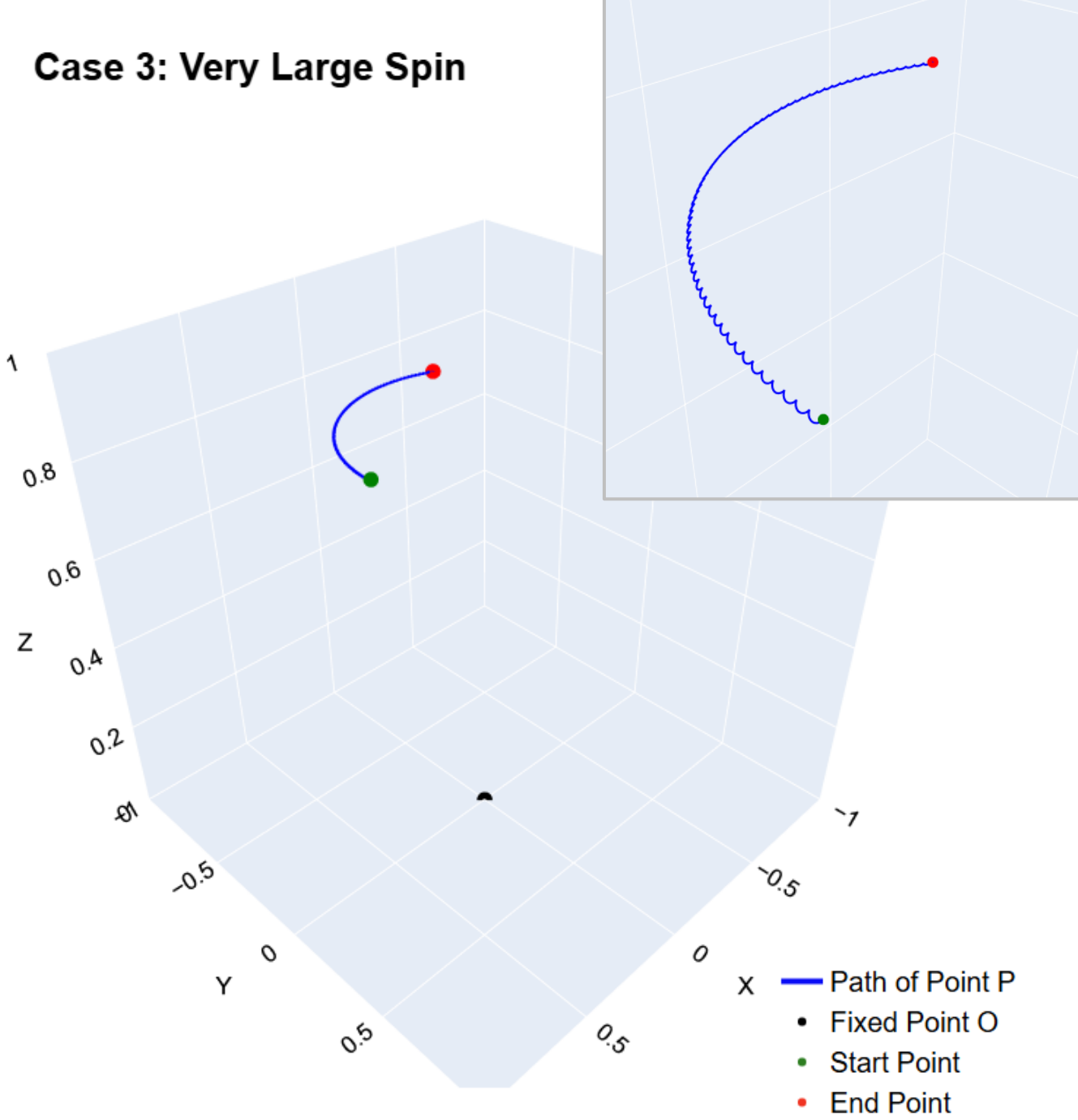




SIMULATION CASE#4

Nutation and Amplitude for the case of large spin





SIMULATION CASE#5

Precession with no Nutation

Condition: $\theta = \theta_0 = \text{const.} \Rightarrow \dot{\theta} = \ddot{\theta} = 0$

From EOM: $I_t\ddot{\theta} + I_a(\dot{\phi} - \dot{\psi}\sin\theta)\dot{\psi}\cos\theta + I_t\dot{\psi}^2\sin\theta\cos\theta + mgl\cos\theta = 0$

Since: $\dot{\phi} = \Omega + \dot{\psi}\sin\theta$

$$I_t\ddot{\theta} + I_a(\Omega + \dot{\psi}\sin\theta - \dot{\psi}\sin\theta)\dot{\psi}\cos\theta + I_t\dot{\psi}^2\sin\theta\cos\theta + mgl\cos\theta = 0$$

Apply conditions: $I_a\Omega\dot{\psi}\cos\theta + I_t\dot{\psi}^2\sin\theta\cos\theta + mgl\cos\theta = 0$

Quadratic Eq.: $(I_t\sin\theta)\dot{\psi}^2 + (I_a\Omega)\dot{\psi} + mgl = 0$

$$\dot{\psi} = \frac{-I_a\Omega \pm \sqrt{(I_a\Omega)^2 - 4(I_t\sin\theta)(mgl)}}{2I_t\sin\theta} \qquad Let:\ b = \frac{I_a\Omega}{I_t} \text{ and } c = \frac{mgl}{I_t}$$

$$\dot{\psi} = \frac{-bI_t \pm \sqrt{b^2I_t^2 - 4(I_t\sin\theta)I_tc}}{2I_t\sin\theta} = \frac{-bI_t \pm \sqrt{b^2I_t^2\left(1 - \frac{4c\sin\theta}{b^2}\right)}}{2I_t\sin\theta} = -\frac{b}{2\sin\theta}\left[1 + \sqrt{1 - \frac{4c\sin\theta}{b^2}}\right]$$

SIMULATION CASE#5

Precession with no Nutation

From Textbook

$$\dot{\psi} = \frac{-b}{2\sin\theta}\left[1 \pm \sqrt{1 - \frac{2c\sin\theta}{b^2}}\right] \qquad (8\text{–}195)$$

It is interesting to note that two steady precession rates are possible, provided that the values of θ and $b = I_a\Omega/I_t$ are such that the square root

My Derivation

$$\dot{\psi} = \frac{-b}{2\sin\theta}\left[1 + \sqrt{1 - \frac{4c\sin\theta}{b^2}}\right]$$

Consider the case in which $\Omega^2 \gg 4I_t mgl/I_a^2$

$$\dot{\psi} = \frac{-b}{2\sin\theta}\left[1 \pm \left(1 - \frac{c\sin\theta}{b^2}\right)\right]$$

$$\dot{\psi}_1 = -\frac{b}{\sin\theta} + \frac{c}{2b}$$

$$\dot{\psi}_2 = \frac{c}{2b}$$

from which we obtain the following possible rates of uniform precession:

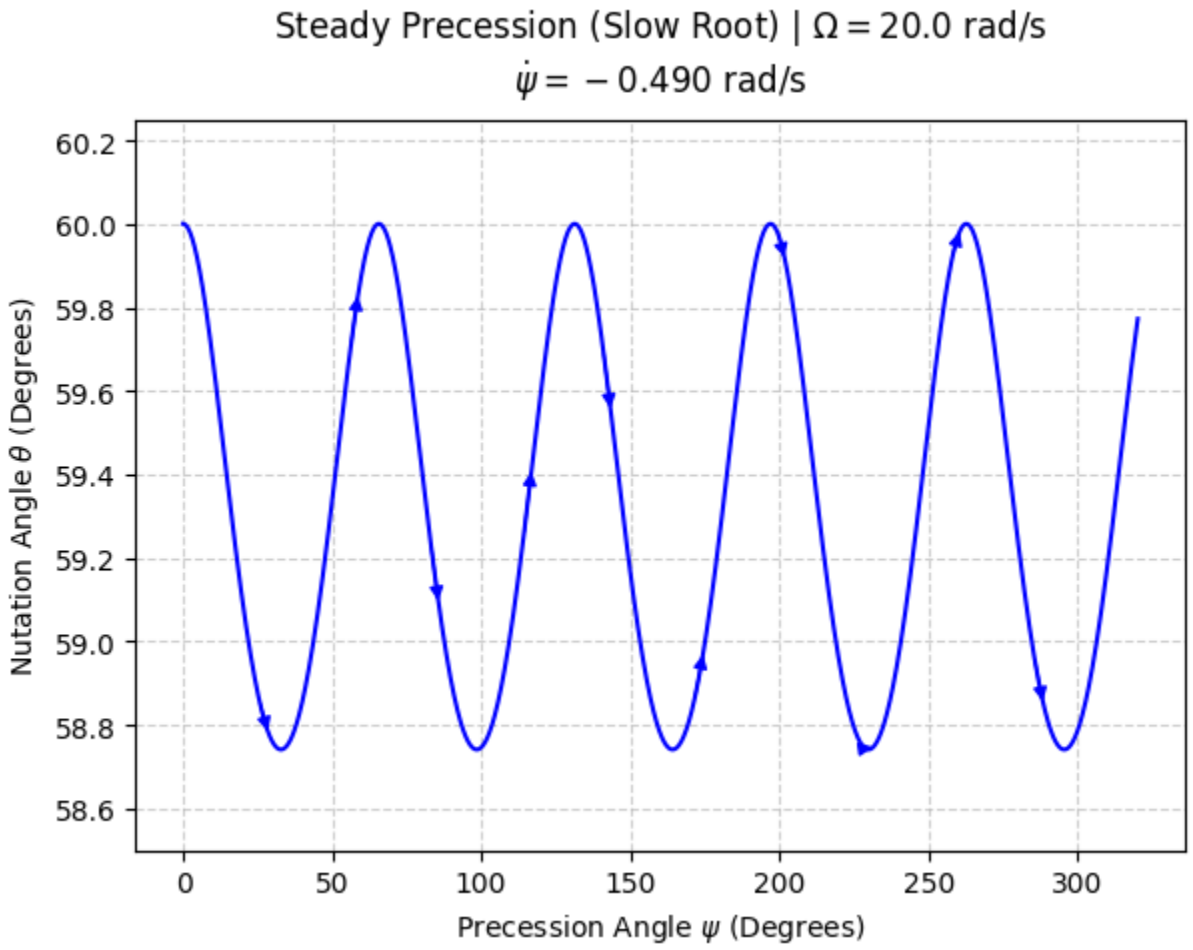
$$(\dot{\psi})_1 = \frac{-c}{2b} = -\frac{mgl}{I_a\Omega}$$

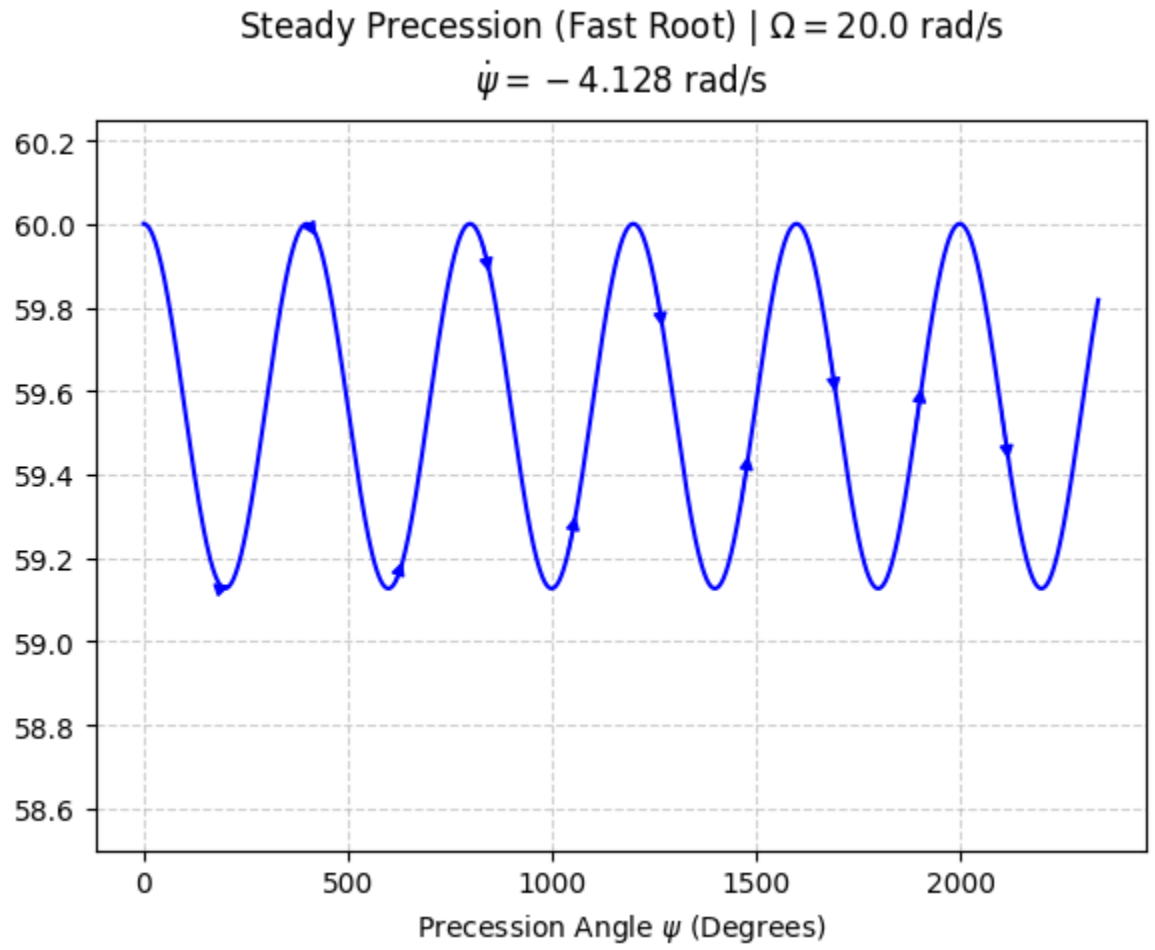
$$(\dot{\psi})_2 = \frac{-b}{\sin\theta} = -\frac{I_a\Omega}{I_t\sin\theta} \qquad (8\text{–}197)$$

SIMULATION CASE#5

Precession with no Nutation

$$\theta(0) = 60°, \qquad \dot{\psi}(0) = 0 \ \text{rad/s}, \qquad \Omega = 20 \ \text{rad/s}$$

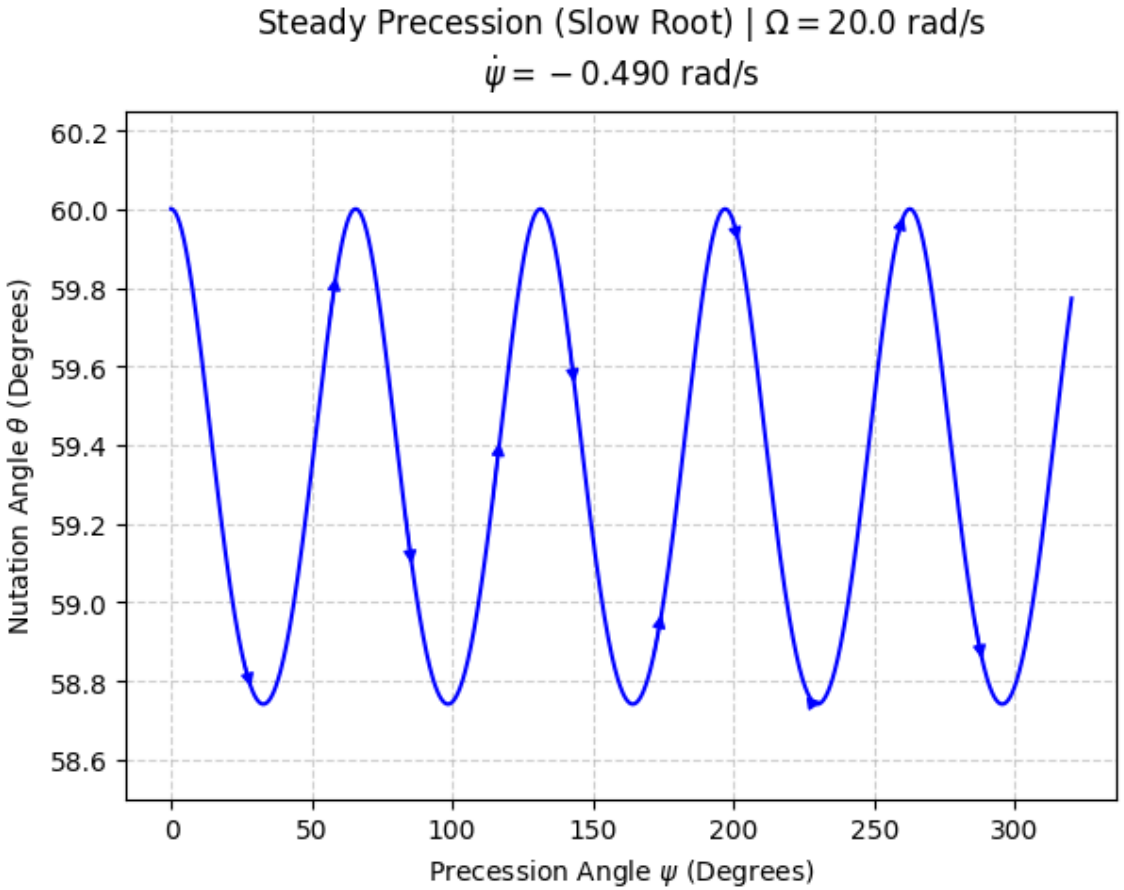


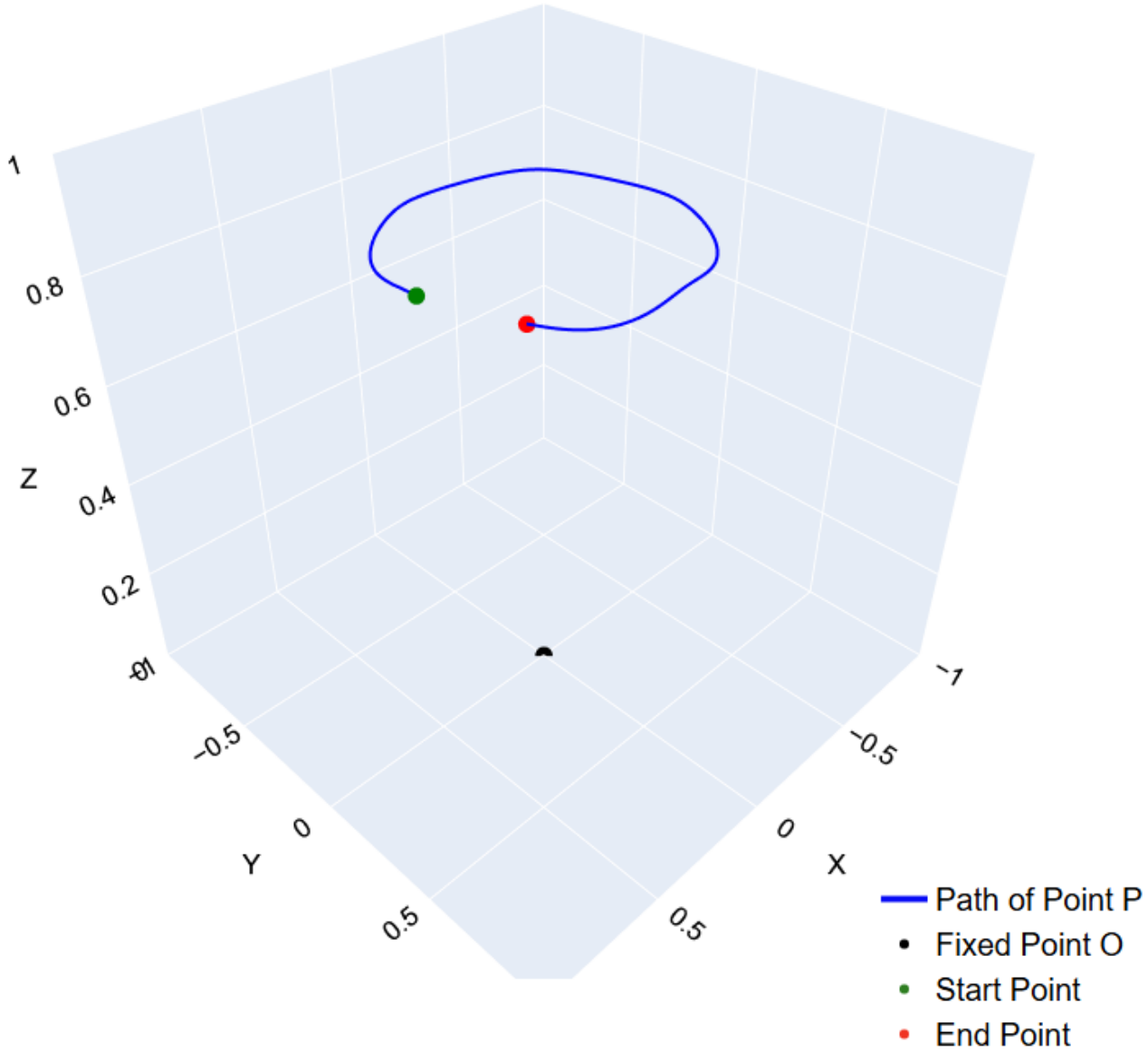


SIMULATION CASE#5

Precession with no Nutation

Slow root: $\dot{\psi} = -0.49$

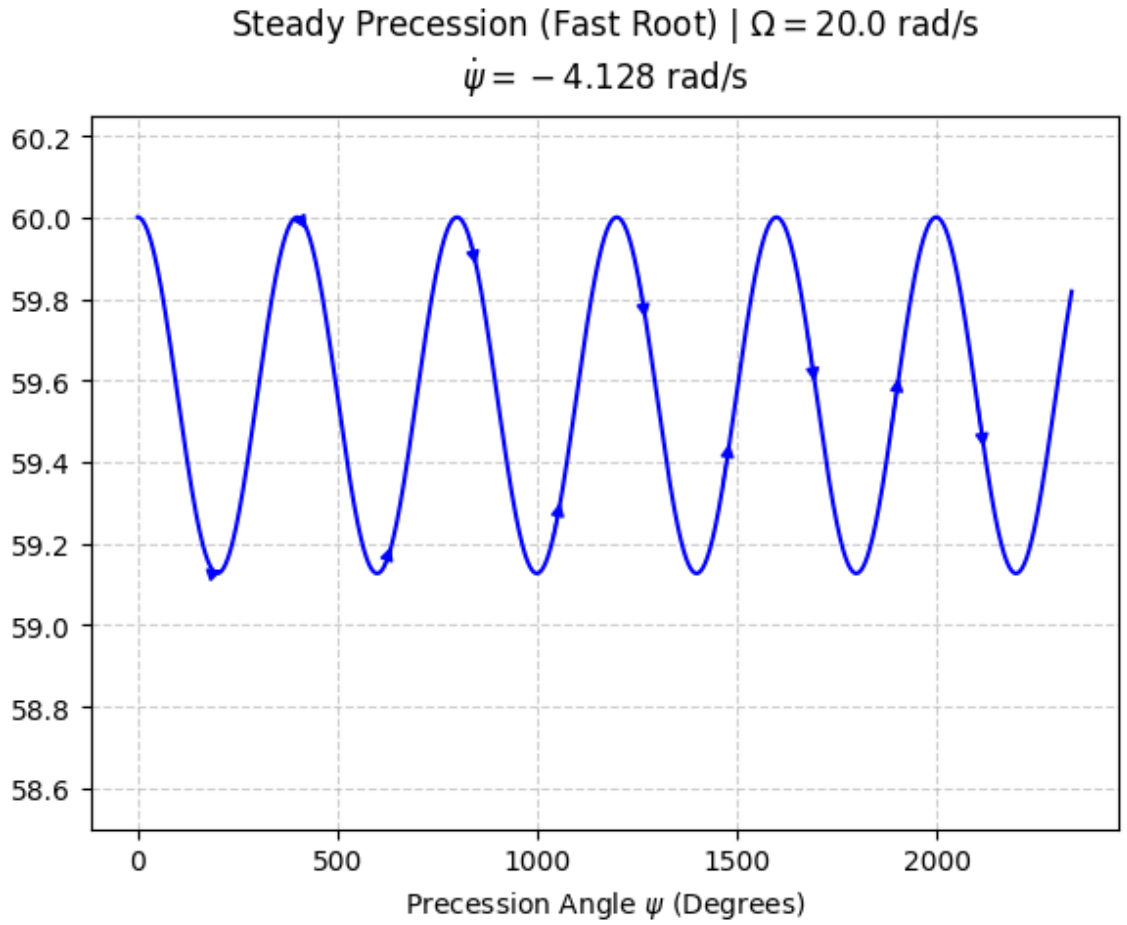


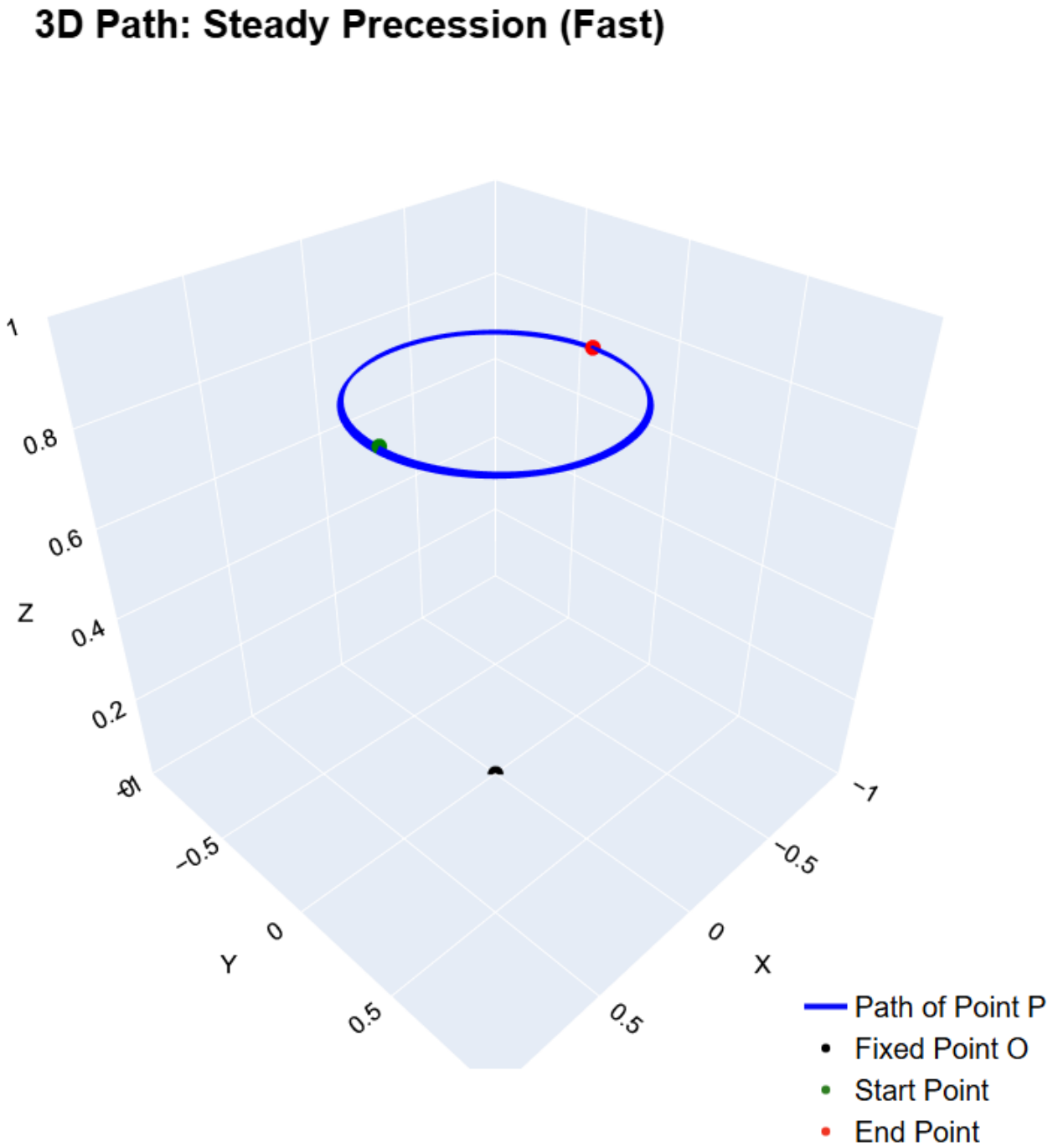


SIMULATION CASE#5

Precession with no Nutation

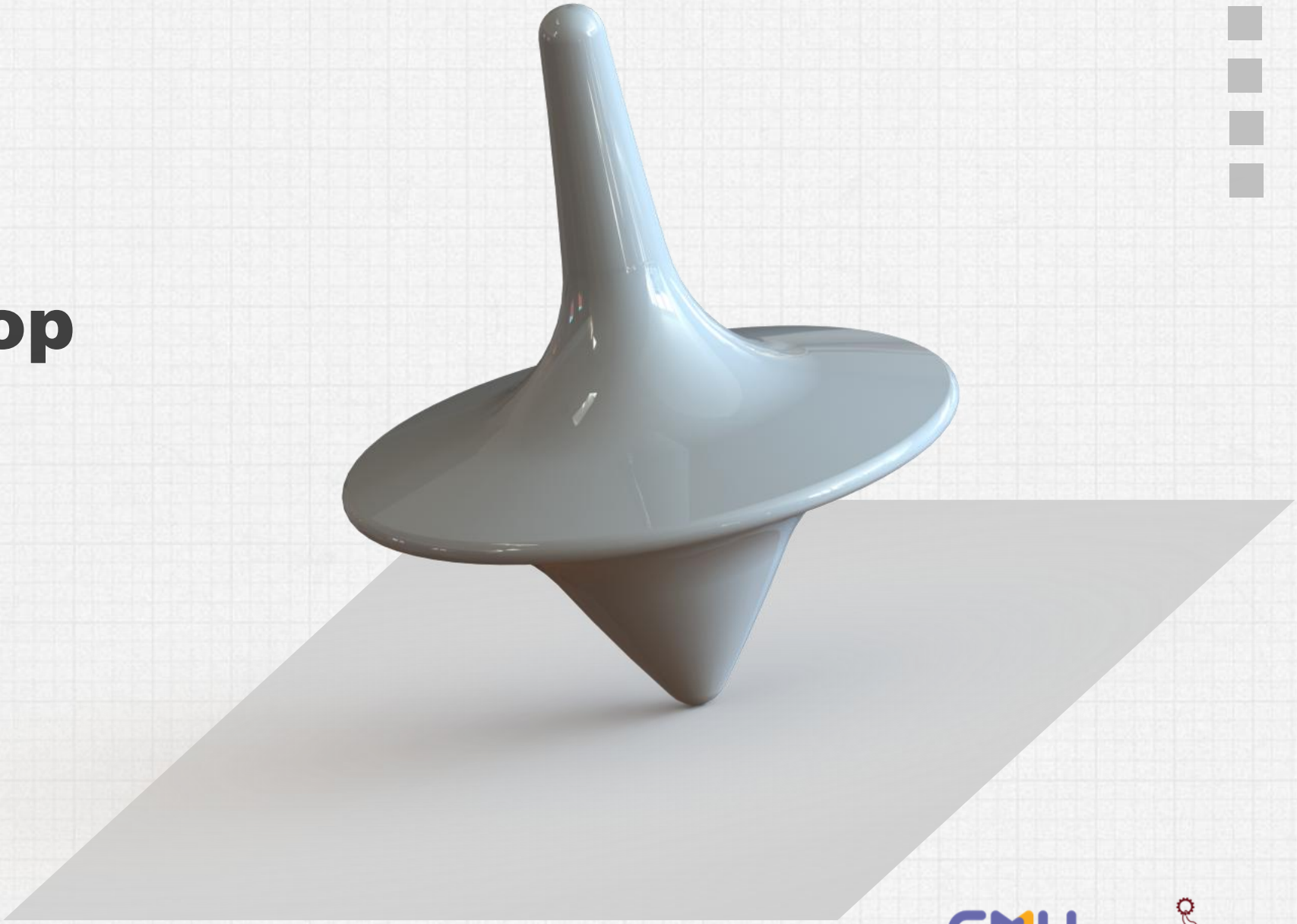
Fast root: $\dot{\psi} = -4.128$





Analytical Dynamics

Final Project
Motion of a Top



THANK YOU

Wongsakorn Wongchompu
Student ID 680831027